Dorey's Resource Book

Differentiation

Derivatives of exponential, logarithmic and trigonometric functions

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1 The derivative of $e^{f(x)}$

ANCHOR QUESTION

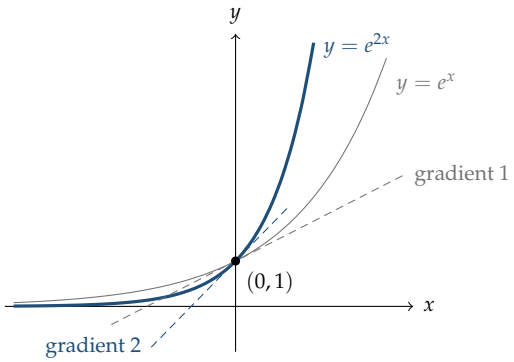
Differentiate $y = e^{x^2-5x}$.

SYLLABUS MAV-12-04

- Apply the rules of differentiation to find the derivative of $f(x) = ke^{ax}$, where k and a are constants
- Use the chain rule to prove $\frac{d}{dx}\left(e^{ax+b}\right) = ae^{ax+b}$, where a and b are constants and $a \neq 0$
- Use the chain rule to differentiate functions of the form $e^{f(x)}$

NOTES

- $\frac{d}{dx}\left(e^x\right) = e^x$.
- $\frac{d}{dx}\left(e^{f(x)}\right) = f'(x)\,e^{f(x)}$. This is on the reference sheet.
- The power stays exactly the same. Multiply by the derivative of the power.
- Special case: $\frac{d}{dx}\left(e^{ax+b}\right) = a\,e^{ax+b}$, and $\frac{d}{dx}\left(ke^{ax}\right) = ka\,e^{ax}$.
- Proof: let $u = ax + b$, so $y = e^u$. Then $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = e^u \times a = ae^{ax+b}$.
- x in the power of e: use this rule. x in the base, as in x^e: use the power rule. No x, as in e^4: a constant, derivative 0.
- Rewrite first with index laws: $\frac{1}{e^{kx}} = e^{-kx}$, $e^a e^b = e^{a+b}$, $\sqrt{e^x} = e^{x/2}$.
- $e^{f(x)} > 0$ for every x, so $f'(x)\,e^{f(x)} = 0$ only when $f'(x) = 0$.



WORKED EXAMPLE

Differentiate $y = e^{3x^2-x}$.

$f(x) = 3x^2 - x$, so $f'(x) = 6x - 1$. the power and its derivative

$$\frac{dy}{dx} = f'(x)\,e^{f(x)}$$
$$= (6x - 1)\,e^{3x^2-x}$$

Tier 1 · the derivative of $e^{f(x)}$

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1. For each function, write the power $f(x)$, then find $f'(x)$.

a) e^{9x}

b) e^{2-x}

c) e^{x^4-3x}

2. Differentiate:

a) $y = e^{7x}$

b) $y = e^{x^2+1}$

c) $y = e^{x^3-6x}$

3. Differentiate:

a) $y = e^{1-5x}$

b) $y = e^{\sqrt{x}}$

c) $y = e^4$

Tier 2 · a coefficient in front

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4. Differentiate:

a) $y = 4e^{3x}$

b) $y = -2e^{5x+1}$

c) $y = \frac{1}{2}e^{6x}$

5. Differentiate:

a) $y = 3e^{x^2}$

b) $y = -e^{2x^2-x}$

6. Show that $\dfrac{d}{dx}\left(\frac{1}{4}e^{8x-2}\right) = 2e^{8x-2}$.

Tier 3 · sums and differences

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7. Differentiate:

a) $y = e^{4x} + e^{-4x}$

b) $y = 3e^{2x} - x^3$

8. Differentiate $y = 2e^{-3x} - 5e^{x} + 7x$.

9. Differentiate $y = x^2 - 4x + e^{2-x}$.

Tier 4 · rewrite with index laws first

10. Differentiate:

a) $y = \dfrac{1}{e^x}$

b) $y = \dfrac{4}{e^{2x}}$

11. Differentiate:

a) $y = e^x \cdot e^{3x}$

b) $y = \left(e^{2x}\right)^3$

12. Differentiate:

a) $y = \sqrt{e^{4x}}$

b) $y = (e^x + e^{-x})^2$, by expanding first

Tier 5 · exact values of the derivative

13. If $f(x) = e^{2x+1}$, find the exact value of $f'(0)$.

14. If $f(x) = e^{x^2-4}$, find the exact value of $f'(2)$.

15. Find the gradient of the tangent to $y = 3e^{-x} + e^x$ at the point where $x = \ln 3$.

Tier 6 · the chain rule on the outside

16. Differentiate $y = (e^x + 1)^3$.

17. Differentiate $y = \left(2 - e^{3x}\right)^4$.

18. Differentiate $y = \sqrt{1 + e^{2x}}$.

Tier 7 · show that, then hence

19. Let $y = e^{x^2} \cdot e^{-2x}$.

a) Show that $\dfrac{dy}{dx} = 2(x-1)\,e^{x^2-2x}$.

b) Hence find the exact coordinates of the point where the tangent is horizontal.

20. Let $y = \dfrac{e^{2x^2}}{e^{8x}}$.

a) Show that $\dfrac{dy}{dx} = 4(x-2)\,e^{2x^2-8x}$.

b) Hence find the exact coordinates of the point where the tangent is horizontal.

21. Let $f(x) = e^{x^3-3x}$.

a) Find $f'(x)$ in fully factorised form.

b) Find the exact coordinates of the points where the tangent is horizontal.

Conceptual

22. A student writes $\frac{d}{dx}\left(e^{3x}\right) = 3x\,e^{3x-1}$. Explain the error and give the correct derivative.

23. Use the chain rule to prove that $\frac{d}{dx}\left(e^{ax+b}\right) = ae^{ax+b}$, where a and b are constants.

24. Explain why the graph of $y = 5e^{2x}$ is increasing for every value of x.

25. The curve $y = e^{x^2+kx}$ has a horizontal tangent at $x = 3$. Find the value of k.

26. Let $y = 5e^{2x}$.

(a) Find $\frac{dy}{dx}$. (1 mark)

(b) Show that $\frac{dy}{dx} = 2y$. (1 mark)

(c) Hence find the gradient of the curve at the point where $y = 10$. (1 mark)

Revision

27. Differentiate $y = (3x-2)^4$.

28. Differentiate $y = \sqrt{x} + \frac{2}{x}$.

29. Use the product rule to differentiate $y = x^2(x+3)^3$, giving your answer in factorised form.

30. Use the quotient rule to differentiate $y = \frac{2x+1}{x-3}$.

31. Find the equation of the tangent to $y = x^3 - 2x$ at the point where $x = 1$.

32. Find the gradient of the normal to $y = x^2 + 4x$ at the point where $x = -1$.

33. Simplify $\frac{e^{2x} \times e^{-5x}}{e^x}$.

34. Solve $e^{2x} = 9$, giving the exact answer.

35. Evaluate $\log_2 32$.

36. Simplify $\ln\left(e^3\right) + \ln 1$.

37. Solve $\log_3(x+1) = 2$.

38. Write $\ln\left(x^2(x+1)\right)$ as a sum of logarithms.

39. Find the exact value of $\sin\frac{2\pi}{3}$.

40. Solve $\sin x = \frac{1}{2}$ for $0 \leq x \leq 2\pi$.

41. Convert $\dfrac{5\pi}{6}$ radians to degrees.

42. Simplify $1 - \cos^2\theta$.

43. State the domain of $f(x) = \dfrac{1}{\sqrt{x+2}}$.

44. If $f(x) = 2x + 1$ and $g(x) = x^2$, find $f(g(3))$.

45. Show that $f(x) = x^3 - x$ is an odd function.

46. Solve $x^2 - 6x + 5 < 0$.

47. Find the inverse function of $f(x) = 3x - 4$.

48. Describe the transformations that map $y = f(x)$ onto $y = 2f(x-1)$.

49. Events A and B are independent, with $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \cap B)$.

50. Two fair dice are rolled. Find the probability that the sum is 7.

51. Solve $|x + 3| = 2x$.

52. Find the angle of inclination of the tangent to $y = x^2$ at the point where $x = \frac{1}{2}$.

2 Derivatives of a^x, products and quotients

ANCHOR QUESTION

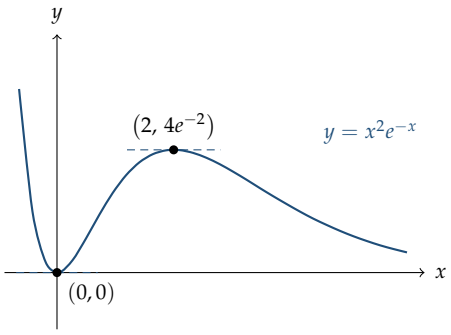
Differentiate $y = x^2e^{3x}$, giving your answer in factorised form.

SYLLABUS MAV-12-04

- Prove and use the formula $\frac{d}{dx}(a^x) = (\ln a)\,a^x$, where a is a constant and $a > 0$
- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\frac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course

NOTES

- Any positive number is a power of e: $a = e^{\ln a}$, so $a^x = e^{x\ln a}$.
- $\frac{d}{dx}(a^x) = (\ln a)\,a^x$. Proof: $\frac{d}{dx}\left(e^{x\ln a}\right) = \ln a \cdot e^{x\ln a} = (\ln a)\,a^x$.
- $\ln a$ is just a number. When $a = e$, $\ln e = 1$ and the rule gives back $\frac{d}{dx}(e^x) = e^x$.
- Product rule: if $y = uv$, then $\frac{dy}{dx} = u'v + uv'$.
- Quotient rule: if $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{u'v - uv'}{v^2}$. Both rules are on the reference sheet.
- After the product rule, both terms share the exponential: take it out as a common factor, e.g. $2xe^{3x} + 3x^2e^{3x} = xe^{3x}(2 + 3x)$.
- An exponential is never 0, so a factorised derivative is 0 only where its other factors are 0.



WORKED EXAMPLE

Differentiate $y = x^2e^{5x}$, giving your answer in factorised form.

$u = x^2,\ u' = 2x; \quad v = e^{5x},\ v' = 5e^{5x}$ — name the parts

$\frac{dy}{dx} = u'v + uv' = 2xe^{5x} + 5x^2e^{5x}$ — product rule

$= xe^{5x}(2 + 5x)$ — common factor xe^{5x}

Tier 1 · the product rule with e

53. Write as a power of e.

a) 9^x

b) 2^{5x}

54. Factorise fully.

a) $3x^2e^{2x} + 2x^3e^{2x}$

b) $e^{3x} - 3xe^{3x}$

55. Differentiate:

a) $y = 6^x$

b) $y = 5 \cdot 2^x$

c) $y = 3^{4x}$

56. Differentiate, giving each answer in factorised form.

a) $y = xe^x$

b) $y = x^2e^{4x}$

57. Differentiate $y = x^4e^x$, giving your answer in factorised form.

Tier 2 · a negative power

58. Differentiate $y = (3x - 1)e^{-x}$, giving your answer in factorised form.

59. Differentiate $y = xe^{-2x}$, giving your answer in factorised form.

60. Differentiate $y = x^2e^{-3x}$, giving your answer in factorised form.

Tier 3 · the quotient rule

61. Differentiate $y = \dfrac{e^{3x}}{x}$.

62. Differentiate $y = \dfrac{x}{e^{2x}}$, simplifying your answer.

63. Differentiate $y = \dfrac{e^x}{x+1}$.

Tier 4 · products with a^x

64. Differentiate $y = x \cdot 2^x$.

65. Differentiate $y = x^2 \cdot 5^x$, giving your answer in factorised form.

66. Differentiate $y = 2^x e^x$.

Tier 5 · quotients of exponentials

67. Differentiate $y = \dfrac{e^x}{e^x + 2}$.

68. Differentiate $y = \dfrac{e^x - 1}{e^x + 1}$.

69. Differentiate $y = \dfrac{e^{2x}}{1 + e^{2x}}$.

Tier 6 · horizontal tangents

70. Find the exact coordinates of the point on $y = xe^{-x}$ where the tangent is horizontal.

71. Find the exact coordinates of the points on $y = x^2 e^x$ where the tangent is horizontal.

72. Find the exact coordinates of the point on $y = \dfrac{e^x}{x}$ where the tangent is horizontal.

Conceptual

73. Prove that $\dfrac{d}{dx}\left(a^x\right) = (\ln a)\, a^x$, where $a > 0$.

74. A student writes $\dfrac{d}{dx}\left(2^x\right) = x \cdot 2^{x-1}$. Explain the error and give the correct derivative.

75. Show that $\dfrac{d}{dx}\left(xe^x\right) = (x+1)e^x$, and hence find the gradient of $y = xe^x$ at the origin.

76. Find the exact gradient of the tangent to $y = xe^{2x}$ at the point where $x = 1$.

77. Let $f(x) = xe^{-x}$.

(a) Show that $f'(x) = (1 - x)e^{-x}$. (1 mark)

(b) Explain why $f(x)$ is increasing for $x < 1$. (1 mark)

(c) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (1 mark)

Revision

78. Differentiate $y = e^{4x-1}$.

79. Differentiate $y = (e^x + 2)^3$.

80. If $f(x) = e^{x^2-1}$, find $f'(1)$.

81. Differentiate $y = \dfrac{1}{e^{2x}}$.

82. Evaluate $\ln\left(e^{-2}\right)$.

83. Solve $3^{x+1} = 27$.

84. Solve $e^x = 5$, giving the exact answer.

85. Evaluate $\log_2 8 + \log_2 4$.

86. Write $\ln\left(x^3 y\right)$ as a sum of logarithms.

87. Differentiate $y = (2x - 5)^3$.

88. Differentiate $y = \sqrt{3x + 1}$.

89. Find the equation of the tangent to $y = x^2 - 4x$ at the point where $x = 3$.

90. Find the gradient of the normal to $y = x^3$ at the point where $x = 1$.

91. Solve $x^2 - x - 6 = 0$.

92. Solve $2x^2 + 5x - 3 = 0$.

93. Find the exact value of $\cos\dfrac{\pi}{3}$.

94. Solve $\tan x = 1$ for $0 \le x \le 2\pi$.

95. Convert $\dfrac{2\pi}{5}$ radians to degrees.

96. State the domain of $y = \ln(x - 3)$.

97. If $f(x) = x^2$ and $g(x) = x + 3$, find $g(f(2))$.

98. Is $f(x) = x^4 + x^2$ even, odd or neither? Give a reason.

99. Solve $|2x + 1| < 5$.

100. Events A and B are mutually exclusive, with $P(A) = 0.3$ and $P(B) = 0.5$. Find $P(A \cup B)$.

101. Find the midpoint of $(-2,\ 5)$ and $(4,\ -1)$.

102. Simplify $\dfrac{x^2 - 9}{x + 3}$.

3 Tangents and normals with exponentials

ANCHOR QUESTION

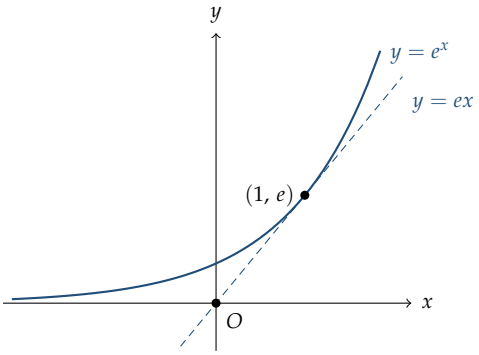
Find the equation of the tangent to $y = e^{2x}$ at the point where $x = 1$.

SYLLABUS MAV-12-04

- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- The gradient of the tangent at $x = a$ is $f'(a)$.
- The normal is perpendicular to the tangent, so its gradient is $-\dfrac{1}{f'(a)}$.
- Find the point with the original function: $y = f(a)$. Find the gradient with the derivative.
- Use $y - y_1 = m(x - x_1)$, then rearrange to $y = mx + c$ or general form.
- Leave e in exact answers: $e^0 = 1$, $e^1 = e$, $e^{\ln k} = k$.



WORKED EXAMPLE

Find the equation of the tangent to $y = e^{4x}$ at the point where $x = 1$.

$x = 1$: $y = e^4$, so the point is $(1, e^4)$. — point from f

$\dfrac{dy}{dx} = 4e^{4x}$, so $m = 4e^4$. — gradient from f'

$y - e^4 = 4e^4(x - 1)$ — point–gradient form

$y = 4e^4x - 3e^4$

Tier 1 · tangents to $y = e^{kx}$

103. Find the exact value.

a) e^{4x} at $x = 1$

b) $3e^{1-x}$ at $x = 1$

c) xe^{3x} at $x = 1$

104. The tangent has gradient m. Find the gradient of the normal.

a) $m = 5e$

b) $m = e^4$

c) $m = -\dfrac{3}{e}$

105. Find the equation of the tangent to $y = e^x$ at the point where $x = 2$.

106. Find the equation of the tangent to $y = e^{5x}$ at the point where $x = 0$.

107. Find the equation of the tangent to $y = 2e^{3x}$ at the point where $x = 1$.

Tier 2 · normals

108. Find the equation of the normal to $y = e^{3x}$ at the point where $x = 0$. Give your answer in general form.

109. Find the equation of the normal to $y = e^x$ at the point where $x = 1$. Give your answer in general form.

110. Find the equation of the normal to $y = e^{-2x}$ at the point where $x = 0$. Give your answer in general form.

Tier 3 · tangents that need the product rule

111. Find the equation of the tangent to $y = xe^{2x}$ at the origin.

112. Find the equation of the tangent to $y = xe^{-x}$ at the point where $x = 2$.

113. Find the equation of the tangent to $y = x^2e^x$ at the point where $x = 1$.

Tier 4 · normals in general form

114. Find the equation of the normal to $y = e^{x^2}$ at the point where $x = -1$. Give your answer in general form.

115. Find the equation of the normal to $y = e^{2x-2}$ at the point where $x = 1$. Give your answer in general form.

116. Find the equation of the normal to $y = e^{x^2-4}$ at the point where $x = 2$. Give your answer in general form.

Tier 5 · show that

117. Show that the tangent to $y = e^{2x}$ at the point where $x = a$ meets the x-axis at $x = a - \frac{1}{2}$.

118. Show that the tangent to $y = e^{-x}$ at the point where $x = a$ meets the x-axis at $x = a + 1$.

119. Show that the tangent to $y = e^x$ at the point where $x = a$ meets the y-axis at $y = (1 - a)e^a$.

Tier 6 · finding the point

120. The tangent to $y = e^{2x}$ at the point where $x = a$ passes through the origin. Find a and the equation of the tangent.

121. Find the exact coordinates of the point on $y = e^x$ where the tangent has gradient 3.

122. Find the exact coordinates of the point on $y = e^{3x}$ where the normal has gradient $-\frac{1}{6}$.

Conceptual

123. Explain why no tangent to $y = e^x$ is horizontal.

124. Show that the tangents to $y = e^x$ and $y = e^{-x}$ at the point $(0, 1)$ are perpendicular.

125. Find where the normal to $y = e^x$ at the point where $x = 0$ crosses the x-axis.

126. A student finds the tangent to $y = e^{2x}$ at $x = 1$ using the point $(1, 2e^2)$. Explain the error.

127. Let $f(x) = e^{2x}$.

(a) Find the equation of the tangent to $y = f(x)$ at the point where $x = 0$. (2 marks)

(b) Find where this tangent meets the x-axis. (1 mark)

Revision

128. Differentiate $y = xe^{3x}$, giving your answer in factorised form.

129. Differentiate $y = 4^x$.

130. Differentiate $y = \dfrac{e^x}{x}$.

131. Differentiate $y = e^{x^3}$.

132. Differentiate $y = (1 - e^x)^2$.

133. Solve $e^{2x} = e^5$.

134. Solve $2^x = 10$, giving the exact answer.

135. Simplify $e^{\ln 7}$.

136. Evaluate $\log_3 81$.

137. Find the gradient of the line $2x - 5y + 1 = 0$.

138. Find the equation of the line through $(-1, 3)$ perpendicular to $y = 2x + 1$. Give your answer in general form.

139. Find the distance between $(1, 2)$ and $(4, 6)$.

140. Differentiate $y = \left(x^2 + 1\right)^4$.

141. Differentiate $y = \dfrac{x}{x - 1}$.

142. Solve $x^2 = 4x$.

143. Solve $x^2 + 2x - 2 = 0$, giving exact answers.

144. Find the exact value of $\sin \dfrac{3\pi}{4}$.

145. Solve $2\cos x = 1$ for $0 \leq x \leq 2\pi$.

146. State the range of $y = e^x + 2$.

147. State the equation of the horizontal asymptote of $y = 3 - e^{-x}$.

148. Find the inverse function of $f(x) = e^x$.

149. Solve $|x - 3| = 5$.

150. If $P(A) = 0.6$, find $P\left(\overline{A}\right)$.

151. Expand $(x + 2)^3$.

152. Simplify $\dfrac{\left(2x^3\right)^2}{4x^2}$.

4 The derivative of $\ln x$ and $\ln f(x)$

ANCHOR QUESTION

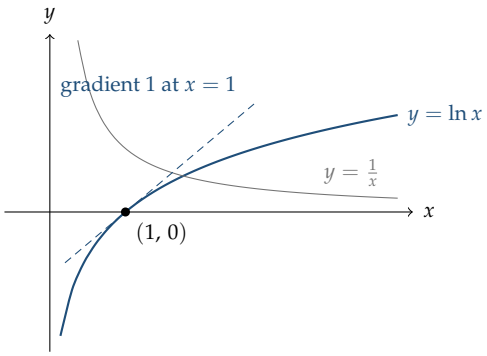
Differentiate $y = \ln\left(x^2 - 3x + 1\right)$.

SYLLABUS MAV-12-04

- Use chain rule on the identity $e^{\ln x} = x$ to prove the formula $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$ for the derivative of the natural logarithm function where $x > 0$
- Use the chain rule to differentiate functions of the form $\ln f(x)$

NOTES

- $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$, for $x > 0$.
- Proof: differentiate both sides of $e^{\ln x} = x$. Then $e^{\ln x} \times \dfrac{d}{dx}(\ln x) = 1$, so $x\,\dfrac{d}{dx}(\ln x) = 1$.
- $\dfrac{d}{dx}\left(\ln f(x)\right) = \dfrac{f'(x)}{f(x)}$: the derivative of the inside, over the inside. This is on the reference sheet.
- $\ln(kx) = \ln k + \ln x$, so $\ln(kx)$ has the same derivative as $\ln x$: $\dfrac{1}{x}$.
- A product, quotient or power inside the log: expand with the log laws first. $\ln(AB) = \ln A + \ln B$, $\ln\dfrac{A}{B} = \ln A - \ln B$, $\ln A^n = n \ln A$.
- A sum or difference inside the log: differentiate it as it is.



WORKED EXAMPLE

Differentiate $y = \ln\left(x^2 + 4x - 1\right)$.

$f(x) = x^2 + 4x - 1$, so $f'(x) = 2x + 4$. the inside and its derivative

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)} = \frac{2x+4}{x^2+4x-1}$$

Tier 1 · the log of a polynomial

153. For each function, write the inside $f(x)$, then find $f'(x)$.

a) $\ln(9x)$

b) $\ln\left(x^2-7\right)$

c) $\ln(5-2x)$

154. Expand using the log laws.

a) $\ln\left(x^3(x+2)\right)$

b) $\ln\left(\dfrac{x-1}{x+5}\right)$

c) $\ln\left((4x+3)^2\right)$

155. Should you use the log laws first? Answer yes or no.

a) $\ln\left((x+1)(x+2)\right)$

b) $\ln\left(2x^3+1\right)$

c) $\ln\left((x-3)^4\right)$

156. Differentiate:

a) $y=\ln\left(x^2+1\right)$

b) $y=\ln\left(x^3-4x\right)$

157. Differentiate $y=\ln\left(2x^2-5x+3\right)$.

Tier 2 · a linear inside

158. Differentiate $y=\ln(6x)$.

159. Differentiate $y=\ln(3x+2)$.

160. Differentiate $y=\ln(1-4x)$.

Tier 3 · coefficients and sums

161. Differentiate $y=5\ln(2x+3)$.

162. Differentiate $y=x^3-2\ln x$.

163. Differentiate $y=\ln\left(x^2+1\right)+e^{2x}$.

Tier 4 · the power law first

164. Differentiate $y = \ln\left((2x+1)^3\right)$.

165. Differentiate $y = \ln\sqrt{4x-1}$.

166. Differentiate $y = \ln\left(x^5\right)$.

Tier 5 · the product and quotient laws first

167. Differentiate $y = \ln\left(x(x+3)\right)$, giving your answer as a single fraction.

168. Differentiate $y = \ln\left(\dfrac{x+2}{x-1}\right)$, giving your answer as a single fraction.

169. Differentiate $y = \ln\left(\dfrac{3x-1}{x+1}\right)$, giving your answer as a single fraction.

Tier 6 · several log laws

170. Differentiate $y = \ln\left(x^3\sqrt{x+1}\right)$, giving your answer as a single fraction.

171. Differentiate $y = \ln\left(\dfrac{(x+1)^2}{x-2}\right)$, giving your answer as a single fraction.

172. Differentiate $y = \ln\sqrt{\dfrac{x+1}{x-1}}$, simplifying fully.

Conceptual

173. Use the identity $e^{\ln x} = x$ to prove that $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$.

174. Show that $\ln(5x)$ and $\ln x$ have the same derivative, and explain why using the log laws.

175. A student writes $\dfrac{d}{dx}\ln\left(x^2+1\right) = \dfrac{1}{x^2+1}$. Explain the error and give the correct derivative.

176. Find the exact gradient of the tangent to $y = \ln\left(x^2+3\right)$ at the point where $x = 1$.

177. Let $f(x) = \ln\left(x^2 - 2x + 5\right)$.

(a) Find $f'(x)$. (1 mark)

(b) Explain why $f(x)$ is defined for all real x. (1 mark)

(c) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (1 mark)

Revision

178. Differentiate $y = e^{3x} + e^{-x}$.

179. Differentiate $y = x^2 e^x$, giving your answer in factorised form.

180. Differentiate $y = 3 \cdot 5^x$.

181. Find the equation of the tangent to $y = e^x$ at the point where $x = 0$.

182. Find the equation of the normal to $y = e^x$ at the point where $x = 0$.

183. An investment grows according to $A = 100e^{0.1t}$. Find $\dfrac{dA}{dt}$ when $t = 0$.

184. Evaluate $e^{2\ln 3}$.

185. Solve $\ln x = 2$.

186. Solve $\ln(x + 1) = 0$.

187. Write $2\ln 3 + \ln 4$ as a single logarithm.

188. Solve $\log_2 x = 5$.

189. Write $\log_5 20$ in terms of natural logarithms.

190. Differentiate $y = \left(3x^2 - 1\right)^5$.

191. Differentiate $y = x\sqrt{x}$.

192. Find the value of x where the curve $y = x^2 - 6x$ has gradient 4.

193. Solve $x^2 - 7x + 12 = 0$.

194. Use the discriminant to state how many real roots $x^2 + x + 3 = 0$ has.

195. Find the exact value of $\tan \dfrac{\pi}{3}$.

196. Solve $\sin x = -\frac{1}{2}$ for $0 \leq x \leq 2\pi$.

197. State the domain of $y = \sqrt{4 - x}$.

198. State the range of $y = \ln x$.

199. Solve $3x - 2 > 7$.

200. If $f(x) = 2x - 1$, find $f^{-1}(5)$.

201. Events A and B are independent, with $P(A) = 0.5$ and $P(B) = 0.4$. Find $P(A \cup B)$.

202. Simplify $\dfrac{x^2 - 4x}{x^2 - 16}$.

5 $\log_a x$, combined rules and tangents

ANCHOR QUESTION

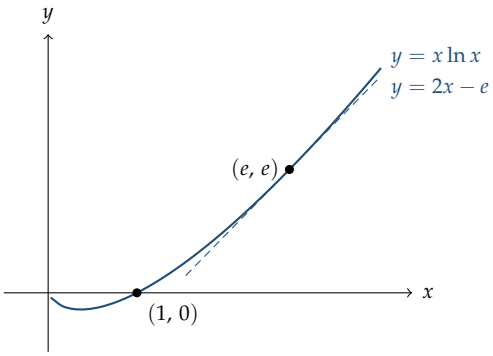
Differentiate $y = x^2 \ln x$, giving your answer in factorised form.

SYLLABUS MAV-12-04

- Prove and use the formula $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$, where a is a constant and $a > 0$
- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\frac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- Change of base: $\log_a x = \frac{\ln x}{\ln a}$.
- $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$. Proof: $\log_a x = \frac{1}{\ln a} \cdot \ln x$, and $\frac{1}{\ln a}$ is a constant.
- For $\log_a f(x)$: change to $\frac{\ln f(x)}{\ln a}$, then use $\frac{f'(x)}{f(x)}$.
- A power inside the log, as in $\ln(x^3)$: log law first, $3 \ln x$.
- A power on the whole log, as in $(\ln x)^3$: chain rule on the outside, $3(\ln x)^2 \times \frac{1}{x}$. Note $\ln^n x$ means $(\ln x)^n$.
- Tangents and normals at $x = e$: use $\ln e = 1$ and $\ln 1 = 0$ to keep answers exact.



WORKED EXAMPLE

Differentiate $y = x^4 \ln x$, giving your answer in factorised form.

$u = x^4$, $u' = 4x^3$; $\quad v = \ln x$, $v' = \frac{1}{x}$ — name the parts

$\frac{dy}{dx} = 4x^3 \ln x + x^4 \times \frac{1}{x} = 4x^3 \ln x + x^3$ — product rule

$= x^3(4 \ln x + 1)$ — common factor x^3

Tier 1 · products with $\ln x$

203. Write using natural logarithms.

a) $\log_6 x$

b) $\log_3 (x^2 - 1)$

204. Differentiate:

a) $y = \log_4 x$

b) $y = 3\log_5 x$

c) $y = \log_2(3x - 1)$

205. For each function, state whether to use a log law first or the chain rule on the outside.

a) $\ln (x^6)$

b) $(\ln x)^6$

c) $\ln^2(3x)$

d) $\ln ((x+4)^3)$

206. Differentiate $y = x \ln x$.

207. Differentiate $y = x^5 \ln x$, giving your answer in factorised form.

Tier 2 · other products with logs

208. Differentiate $y = e^x \ln x$.

209. Differentiate $y = x^2 \ln(3x)$, giving your answer in factorised form.

210. Differentiate $y = \sqrt{x}\, \ln x$, giving your answer as a single fraction.

Tier 3 · the quotient rule with logs

211. Differentiate $y = \dfrac{\ln x}{x^2}$.

212. Differentiate $y = \dfrac{x}{\ln x}$.

213. Differentiate $y = \dfrac{\ln x}{x+1}$, giving your answer as a single fraction.

Tier 4 · powers of $\ln x$

214. Differentiate $y = (\ln x)^2$.

215. Differentiate $y = (\ln x)^4$.

216. Differentiate $y = \sqrt{\ln x}$.

Tier 5 · tangents with logs

217. Find the equation of the tangent to $y = \ln x$ at the point where $x = 1$.

218. Find the equation of the tangent to $y = \ln(3x)$ at the point where $x = \frac{1}{3}$.

219. Find the equation of the tangent to $y = 2\ln x$ at the point where $x = e$.

Tier 6 · normals and combined rules

220. Find the equation of the normal to $y = \ln x$ at the point where $x = e$.

221. Find the equation of the tangent to $y = x^2 \ln x$ at the point where $x = 1$.

222. Find the equation of the normal to $y = x\ln x$ at the point where $x = 1$.

Conceptual

223. Prove that $\dfrac{d}{dx}(\log_a x) = \dfrac{1}{x\ln a}$.

224. Find $\dfrac{d}{dx}\ln(x^2)$ and $\dfrac{d}{dx}(\ln x)^2$, and explain why they are different.

225. Show that the tangent to $y = \ln x$ at the point where $x = e$ passes through the origin.

226. Find the exact coordinates of the point on $y = x\ln x$ where the tangent is horizontal.

227. Let $f(x) = \dfrac{\ln x}{x}$, for $x > 0$.

(a) Show that $f'(x) = \dfrac{1 - \ln x}{x^2}$. (1 mark)

(b) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (2 marks)

Revision

228. Differentiate $y = \ln\left(x^2 + 7\right)$.

229. Differentiate $y = \ln\left(\dfrac{x+1}{x-1}\right)$, giving your answer as a single fraction.

230. Differentiate $y = xe^{-x}$, giving your answer in factorised form.

231. Differentiate $y = 10^x$.

232. Differentiate $y = e^{\sqrt{x}}$.

233. Find the equation of the tangent to $y = e^{2x}$ at the point where $x = 0$.

234. Show that $\dfrac{d}{dx}\left(e^{2x} - e^{-2x}\right) = 2\left(e^{2x} + e^{-2x}\right)$.

235. Solve $e^{2x} - 3e^x + 2 = 0$.

236. Solve $\ln(2x - 1) = 0$.

237. Simplify $\ln\left(e^x\right) + e^{\ln x}$.

238. Evaluate $\log_4 2$.

239. Write $\ln 18 - \ln 2 + \ln 5$ as a single logarithm.

240. Differentiate $y = \dfrac{4x+1}{2x-3}$.

241. Differentiate $y = \dfrac{6}{\sqrt{x}}$.

242. Find the equation of the normal to $y = x^2$ at the point $(1, 1)$. Give your answer in general form.

243. Solve $x^3 - 4x = 0$.

244. Find the values of k for which $x^2 + kx + 9 = 0$ has exactly one real root.

245. Find the exact value of $\cos\dfrac{5\pi}{6}$.

246. Solve $2\sin^2 x = 1$ for $0 \leq x \leq 2\pi$.

247. State the period of $y = \sin(3x)$.

248. State the domain of $y = \dfrac{1}{x-2}$.

249. Solve $|x + 1| \geq 3$.

250. If $f(x) = \sqrt{x}$ and $g(x) = x^2 + 9$, find $f(g(4))$.

251. If $P(A) = 0.3$ and $P(B \mid A) = 0.5$, find $P(A \cap B)$.

252. Factorise $x^3 - 8$.

6 Derivatives of $\sin x$, $\cos x$ and $\tan x$

ANCHOR QUESTION

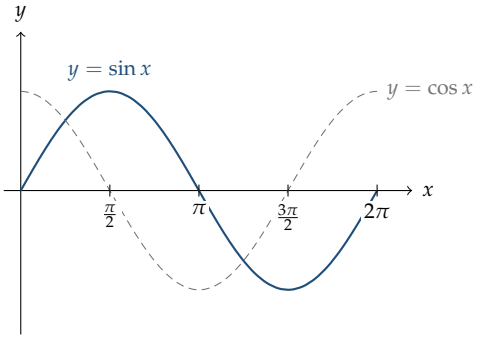
Differentiate $y = \sin\left(x^2 + 2x\right)$.

SYLLABUS MAV-12-04

- Determine the formulas $\frac{d}{dx}(\sin x) = \cos x$ and $\frac{d}{dx}(\cos x) = -\sin x$ informally by using the graphs of $y = \sin x$ and $y = \cos x$ and verify with graphing applications
- Use the rules of differentiation to show that $\frac{d}{dx}(\tan x) = \sec^2 x$
- Use the chain rule to differentiate functions of the form $\sin f(x)$, $\cos f(x)$ and $\tan f(x)$

NOTES

- $\frac{d}{dx}(\sin x) = \cos x$ and $\frac{d}{dx}(\cos x) = -\sin x$. Angles must be in radians.
- From the graph of $y = \sin x$: the gradient is 0 at the turning points $x = \frac{\pi}{2}$, $\frac{3\pi}{2}$, is 1 at $x = 0$ and -1 at $x = \pi$. That is the graph of $y = \cos x$.
- $\frac{d}{dx}(\tan x) = \sec^2 x$, where $\sec x = \frac{1}{\cos x}$. Proof: write $\tan x = \frac{\sin x}{\cos x}$ and use the quotient rule with $\sin^2 x + \cos^2 x = 1$.
- Chain rule: $\frac{d}{dx}\sin f(x) = f'(x)\cos f(x)$, $\frac{d}{dx}\cos f(x) = -f'(x)\sin f(x)$, $\frac{d}{dx}\tan f(x) = f'(x)\sec^2 f(x)$.
- The angle stays the same; multiply by the derivative of the angle.



WORKED EXAMPLE

Differentiate $y = \cos\left(x^2 + 4x\right)$.

$f(x) = x^2 + 4x$, so $f'(x) = 2x + 4$. — the angle and its derivative

$\frac{dy}{dx} = -f'(x)\sin f(x)$ — cos gives − sin

$= -(2x + 4)\sin\left(x^2 + 4x\right)$

Tier 1 · $\sin f(x)$

253. For each function, write the angle $f(x)$, then find $f'(x)$.

a) $\sin(9x)$

b) $\cos\left(x^2 - 1\right)$

c) $\tan(5 - 2x)$

254. Differentiate:

a) $y = \sin x + \cos x$

b) $y = 4\tan x$

c) $y = x^2 - 3\cos x$

255. Differentiate $y = \sin\left(x^2 + 3x\right)$.

256. Differentiate $y = \sin\left(x^3\right)$.

257. Differentiate $y = \sin(5x - 2)$.

Tier 2 · $\cos f(x)$

258. Differentiate $y = \cos\left(x^2 + 1\right)$.

259. Differentiate $y = \cos 4x$.

260. Differentiate $y = \cos(2 - 3x)$.

Tier 3 · $\tan f(x)$

261. Differentiate $y = \tan 6x$.

262. Differentiate $y = \tan\left(x^2\right)$.

263. Differentiate $y = \tan(1 - x)$.

Tier 4 · coefficients and sums

264. Differentiate $y = 5\sin 3x + 2\cos 4x$.

265. Differentiate $y = 2\tan\dfrac{x}{2}$.

266. Differentiate $y = \cos(\pi x) - \sin(\pi x)$.

Tier 5 · exact gradients

267. Find the exact gradient of the tangent to $y = \cos 2x$ at the point where $x = \dfrac{\pi}{12}$.

268. Find the exact gradient of the tangent to $y = \tan x$ at the point where $x = \dfrac{\pi}{4}$.

269. Find the exact gradient of the tangent to $y = \sin 3x$ at the point where $x = \dfrac{\pi}{9}$.

Tier 6 · horizontal tangents

270. Find the values of x, for $0 \leq x \leq 2\pi$, where the tangent to $y = \sin x - \cos x$ is horizontal.

271. Find the values of x, for $0 \leq x \leq 2\pi$, where the tangent to $y = x + 2\cos x$ is horizontal.

272. Find the values of x, for $0 \leq x \leq \pi$, where the tangent to $y = \sin 2x$ is horizontal.

Conceptual

273. Use the quotient rule to show that $\dfrac{d}{dx}(\tan x) = \sec^2 x$.

274. Use the graph of $y = \sin x$ to explain why its gradient is 1 at $x = 0$, 0 at $x = \dfrac{\pi}{2}$ and -1 at $x = \pi$, and why this matches $y = \cos x$.

275. A student writes $\dfrac{d}{dx}(\cos 3x) = -\sin 3x$. Explain the error and give the correct derivative.

276. Explain why the graph of $y = \tan x$ has no horizontal tangents.

277. Let $f(x) = \sin x + \sqrt{3}\cos x$, for $0 \leq x \leq 2\pi$.

(a) Find $f'(x)$. (1 mark)

(b) Find the values of x where the tangent to $y = f(x)$ is horizontal. (2 marks)

Revision

278. Differentiate $y = \ln(5x - 2)$.

279. Differentiate $y = x \ln x$.

280. Differentiate $y = \log_3 x$.

281. Differentiate $y = (\ln x)^2$.

282. Differentiate $y = e^{-3x}$.

283. Differentiate $y = x^3 e^{2x}$, giving your answer in factorised form.

284. Find the equation of the tangent to $y = \ln x$ at the point where $x = e^2$.

285. Convert 225° to radians.

286. Convert $\dfrac{7\pi}{6}$ radians to degrees.

287. Find the exact value of $\sin \dfrac{5\pi}{4}$.

288. Find the exact value of $\tan \dfrac{2\pi}{3}$.

289. Solve $\cos x = -\frac{1}{2}$ for $0 \leq x \leq 2\pi$.

290. Solve $\tan x = \sqrt{3}$ for $0 \leq x \leq 2\pi$.

291. Simplify $\dfrac{1 - \cos^2 x}{\sin x}$.

292. State the amplitude and period of $y = 3\cos 2x$.

293. Find the exact length of an arc of a circle with radius 6 cm that subtends an angle of $\dfrac{\pi}{3}$ at the centre.

294. Find the exact area of a sector with radius 4 cm and angle $\dfrac{\pi}{2}$.

295. Solve $2^{x+1} = 32$.

296. Solve $\ln x + \ln 2 = \ln 10$.

297. Differentiate $y = \left(x^2 - 3\right)^4$.

298. Find the equation of the normal to $y = x^3$ at the point $(1, 1)$. Give your answer in general form.

299. Solve $x^2 + 4x - 5 < 0$.

300. Is $f(x) = \sin x$ even, odd or neither? Give a reason.

301. Two fair dice are rolled. Find the probability that both show the same number.

302. Expand $(2x - 1)(x + 3)$.

7 Trig functions with combined rules

ANCHOR QUESTION

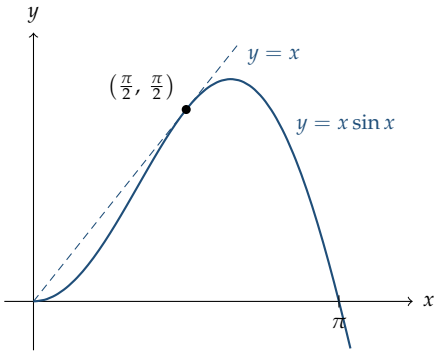
Differentiate $y = x^2 \sin x$, giving your answer in factorised form.

SYLLABUS MAV-12-04

- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\dfrac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- $\sin^n x$ means $(\sin x)^n$: use the chain rule on the outside, e.g. $\dfrac{d}{dx}\sin^2 x = 2\sin x \cos x$.
- Product rule $u'v + uv'$ and quotient rule $\dfrac{u'v - uv'}{v^2}$ work with trig functions exactly as before. Name u, v, u', v' first.
- Take out common factors, e.g. $2x\sin x + x^2\cos x = x(2\sin x + x\cos x)$.
- Trig inside e or ln: $\dfrac{d}{dx}e^{\sin x} = \cos x\, e^{\sin x}$ and $\dfrac{d}{dx}\ln(\cos x) = \dfrac{-\sin x}{\cos x} = -\tan x$.
- For tangents, use exact values: $\sin\dfrac{\pi}{2} = 1$, $\cos\dfrac{\pi}{2} = 0$, $\cos\pi = -1$.



WORKED EXAMPLE

Differentiate $y = x^3\cos x$, giving your answer in factorised form.

$u = x^3$, $u' = 3x^2$; $\quad v = \cos x$, $v' = -\sin x$ — name the parts

$\dfrac{dy}{dx} = 3x^2\cos x - x^3\sin x$ — product rule

$= x^2(3\cos x - x\sin x)$ — common factor x^2

Tier 1 · products with $\sin x$, $\cos x$, $\tan x$

303. Rewrite with the power outside a bracket.

a) $\cos^3 x$

b) $\sin^2 4x$

304. Differentiate $y = x \sin x$.

305. Differentiate $y = x^2 \cos x$, giving your answer in factorised form.

306. Differentiate $y = x^3 \tan x$, giving your answer in factorised form.

307. Differentiate $y = \sqrt{x} \sin x$.

Tier 2 · products with a chain inside

308. Differentiate $y = x \sin 3x$.

309. Differentiate $y = x^2 \cos 2x$, giving your answer in factorised form.

310. Differentiate $y = \sin x \cos x$.

Tier 3 · powers of trig functions

311. Differentiate $y = \cos^2 x$.

312. Differentiate $y = \sin^3 x$.

313. Differentiate $y = \tan^2 x$.

Tier 4 · the quotient rule with trig

314. Differentiate $y = \dfrac{\cos x}{x}$.

315. Differentiate $y = \dfrac{x}{\sin x}$.

316. Show that $\dfrac{d}{dx}\left(\dfrac{\sin x}{1+\cos x}\right) = \dfrac{1}{1+\cos x}$.

Tier 5 · trig with e and $\ln$

317. Differentiate $y = e^x \sin x$, giving your answer in factorised form.

318. Differentiate $y = \ln(\sin x)$.

319. Differentiate $y = e^{\cos x}$.

Tier 6 · tangents and normals

320. Find the equation of the tangent to $y = x\cos x$ at the point where $x = \pi$.

321. Find the equation of the tangent to $y = \sin^2 x$ at the point where $x = \dfrac{\pi}{4}$.

322. Find the equation of the normal to $y = e^x \cos x$ at the point where $x = 0$.

Tier 7 · show that

323. Show that $\dfrac{d}{dx}\left(\dfrac{\cos x}{1-\sin x}\right) = \dfrac{1}{1-\sin x}$.

324. Show that $\dfrac{d}{dx}(x\sin x + \cos x) = x\cos x$.

325. Show that $\dfrac{d}{dx}\left[e^{-x}(\sin x + \cos x)\right] = -2e^{-x}\sin x$.

Conceptual

326. Show that $\dfrac{d}{dx}\left(\sin^2 x + \cos^2 x\right) = 0$, and explain why this makes sense.

327. Show that $\dfrac{d}{dx}\ln(\sin x) = \cot x$.

328. A student writes $\dfrac{d}{dx}(x\sin x) = \cos x$. Explain the error and give the correct derivative.

329. Find the exact gradient of the tangent to $y = x^2 \sin x$ at the point where $x = \pi$.

330. Let $f(x) = e^{-x}\sin x$, for $0 \le x \le 2\pi$.

(a) Show that $f'(x) = e^{-x}(\cos x - \sin x)$. (1 mark)

(b) Find the values of x where the tangent to $y = f(x)$ is horizontal. (2 marks)

Revision

331. Differentiate $y = \sin\left(x^2\right)$.

332. Differentiate $y = \tan 5x$.

333. Differentiate $y = \cos(3x + 1)$.

334. Find the exact gradient of the tangent to $y = \cos x$ at the point where $x = \dfrac{\pi}{6}$.

335. Differentiate $y = e^{x^2}$.

336. Differentiate $y = \ln\left(x^3 + 1\right)$.

337. Differentiate $y = 2^{3x}$.

338. Differentiate $y = \log_{10} x$.

339. Differentiate $y = xe^x$, giving your answer in factorised form.

340. Differentiate $y = \dfrac{\ln x}{x}$.

341. Find the equation of the tangent to $y = e^x$ at the point where $x = 1$.

342. Find the equation of the normal to $y = \ln x$ at the point where $x = 1$.

343. Find the exact value of $\cos \dfrac{7\pi}{6}$.

344. Solve $\sin 2x = 0$ for $0 \leq x \leq \pi$.

345. Solve $2\cos x + \sqrt{3} = 0$ for $0 \leq x \leq 2\pi$.

346. Simplify $\tan^2 x + 1$.

347. State the range of $y = 2\sin x + 1$.

348. State the period of $y = \tan 2x$.

349. Solve $e^x = \dfrac{1}{e}$.

350. Simplify $\ln\left(e^2 x\right) - \ln x$.

351. Evaluate $\log_2 48 - \log_2 3$.

352. Differentiate $y = (1 - 2x)^5$.

353. Differentiate $y = \dfrac{x^2}{x + 1}$.

354. Solve $x^2 - 2x - 1 = 0$, giving exact answers.

355. Find the gradient of a line perpendicular to $3x + 4y = 12$.

356. If $f(x) = x^2 - 1$, find and simplify $f(a + 1)$.

357. A fair coin is tossed three times. Find the probability of at least one head.

358. Simplify $\sqrt{50} - \sqrt{18}$.

8 Derivatives of the reciprocal trig functions

ANCHOR QUESTION

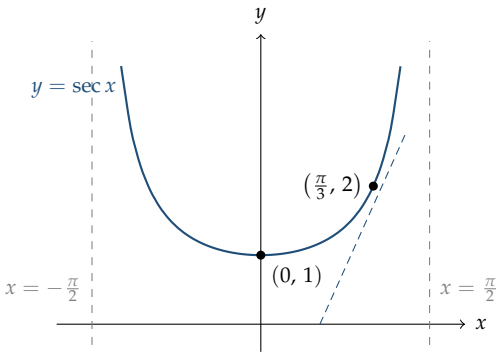
Show that $\dfrac{d}{dx}(\sec x) = \tan x \sec x$.

SYLLABUS MAV-12-04

- Use the rules of differentiation to find the derivatives of cosec x, sec x and cot x
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- $\sec x = \dfrac{1}{\cos x} = (\cos x)^{-1}$, $\operatorname{cosec} x = \dfrac{1}{\sin x} = (\sin x)^{-1}$, $\cot x = \dfrac{\cos x}{\sin x}$.
- $\dfrac{d}{dx}(\sec x) = \tan x \sec x$: chain rule on $(\cos x)^{-1}$.
- $\dfrac{d}{dx}(\operatorname{cosec} x) = -\cot x \operatorname{cosec} x$: chain rule on $(\sin x)^{-1}$.
- $\dfrac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$: quotient rule on $\dfrac{\cos x}{\sin x}$.
- These three are *not* on the reference sheet. The "co" functions get the minus sign.
- With an angle $f(x)$, multiply by $f'(x)$ as usual, e.g. $\dfrac{d}{dx}\sec 2x = 2\tan 2x \sec 2x$.



WORKED EXAMPLE

Show that $\dfrac{d}{dx}(\sec 4x) = 4\tan 4x \sec 4x$.

$y = (\cos 4x)^{-1}$ — write as a power

$\dfrac{dy}{dx} = -(\cos 4x)^{-2} \times (-4\sin 4x)$ — chain rule

$= \dfrac{4\sin 4x}{\cos^2 4x} = 4 \cdot \dfrac{\sin 4x}{\cos 4x} \cdot \dfrac{1}{\cos 4x} = 4\tan 4x \sec 4x$

Tier 1 · $\sec x$ as a power

359. Write as a power of a trig function.

a) $\operatorname{cosec} 5x$

b) $\sec\left(x^3\right)$

360. Find the exact value.

a) $\sec \dfrac{\pi}{6}$

b) $\operatorname{cosec} \dfrac{\pi}{3}$

c) $\cot \dfrac{\pi}{4}$

d) $\operatorname{cosec} \dfrac{\pi}{2}$

361. By writing $\sec 3x = (\cos 3x)^{-1}$, show that $\dfrac{d}{dx}(\sec 3x) = 3\tan 3x \sec 3x$.

362. Differentiate $y = \sec^2 x$.

363. Differentiate $y = 5\sec x$.

Tier 2 · $\operatorname{cosec} x$

364. Differentiate $y = \operatorname{cosec} x + x^2$.

365. Differentiate $y = 2\operatorname{cosec} x$.

366. Show that $\dfrac{d}{dx}(\operatorname{cosec} 2x) = -2\cot 2x \operatorname{cosec} 2x$.

Tier 3 · $\cot x$

367. Differentiate $y = \cot x - x$.

368. Differentiate $y = 4\cot x$.

369. By writing $\cot x = (\tan x)^{-1}$, show that $\dfrac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$.

Tier 4 · sums

370. Differentiate $y = \sec x + \operatorname{cosec} x$.

371. Differentiate $y = 2\cot x - 3\sec x$.

372. Differentiate $y = \tan x + \cot x$.

Tier 5 · the chain rule

373. Differentiate $y = \sec\left(x^2\right)$.

374. Differentiate $y = \cot 3x$.

375. Differentiate $y = \operatorname{cosec}(2x - 1)$.

Tier 6 · the product rule

376. Differentiate $y = x \operatorname{cosec} x$, giving your answer in factorised form.

377. Differentiate $y = x^2 \cot x$, giving your answer in factorised form.

378. Differentiate $y = e^x \sec x$, giving your answer in factorised form.

Tier 7 · tangents and normals

379. Find the equation of the tangent to $y = \sec x$ at the point where $x = \dfrac{\pi}{3}$.

380. Find the gradient of the normal to $y = \operatorname{cosec} x$ at the point where $x = \dfrac{\pi}{4}$.

381. Find the equation of the tangent to $y = \cot x$ at the point where $x = \dfrac{\pi}{2}$.

Conceptual

382. Show that $\dfrac{d}{dx}(\sec x) = \tan x \sec x$.

383. Use the quotient rule to show that $\dfrac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$.

384. Explain why $y = \cot x$ is decreasing wherever it is defined.

385. A student says that since $\sec x = \dfrac{1}{\cos x}$, its derivative is $\dfrac{1}{-\sin x}$. Explain the error.

386. Let $f(x) = \sec x$, for $-\dfrac{\pi}{2} < x < \dfrac{\pi}{2}$.

(a) Show that $f'(x) = \dfrac{\sin x}{\cos^2 x}$. (1 mark)

(b) Find the point where the tangent to $y = f(x)$ is horizontal. (1 mark)

(c) Explain why $f'(x) > 0$ for $0 < x < \dfrac{\pi}{2}$. (1 mark)

Revision

387. Differentiate $y = x \sin x$.

388. Differentiate $y = \cos^2 x$.

389. Differentiate $y = \tan 3x$.

390. Differentiate $y = e^{\sin x}$.

391. Differentiate $y = \dfrac{\sin x}{x}$.

392. Differentiate $y = \ln\left(x^2 + 1\right)$.

393. Differentiate $y = 5^x$.

394. Differentiate $y = x^2 e^{-x}$, giving your answer in factorised form.

395. Find the equation of the tangent to $y = \sin x$ at the point where $x = \pi$.

396. Find the exact gradient of the tangent to $y = \tan x$ at the point where $x = \dfrac{\pi}{3}$.

397. Solve $\sin x = \cos x$ for $0 \leq x \leq 2\pi$.

398. Find the exact value of $\sin \dfrac{4\pi}{3}$.

399. Convert 300° to radians.

400. Simplify $\sin x \operatorname{cosec} x$.

401. Simplify $1 + \cot^2 x$.

402. Solve $2 \sin x - 1 = 0$ for $0 \leq x \leq 2\pi$.

403. Solve $\ln(x - 2) = \ln 5$.

404. Solve $3e^{2x} = 12$, giving the exact answer.

405. Evaluate $\log_3 5 + \log_3 \frac{1}{5}$.

406. Differentiate $y = 3x^4 - \dfrac{2}{x}$.

407. Differentiate $y = \sqrt{x^2 + 4}$.

408. Find the values of x where the curve $y = x^3 - 12x$ has a horizontal tangent.

409. Solve $x^2 > 9$.

410. The line $y = 2x + c$ passes through $(3, 1)$. Find c.

411. Solve simultaneously: $y = x^2$ and $y = x + 2$.

412. State the domain of $y = \ln\left(x^2\right)$.

413. Two fair dice are rolled. Find the probability that the sum is 10.

414. Simplify $\left(3a^2b\right)^3$.

9 Mixed differentiation: choosing the rule

ANCHOR QUESTION

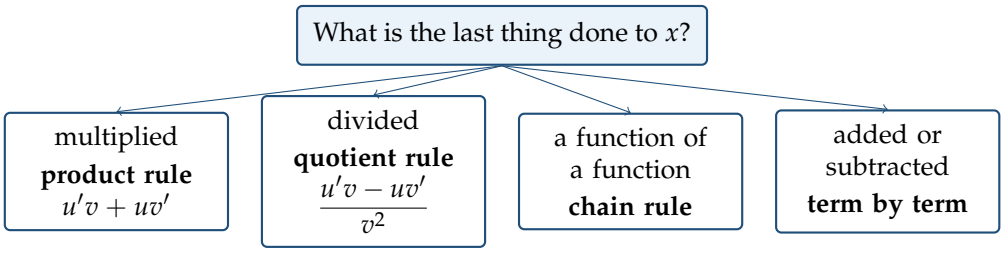
Differentiate $y = e^{2x}\sin 3x$, giving your answer in factorised form.

SYLLABUS MAV-12-04

- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\dfrac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- Standard derivatives: $e^x \to e^x$, $\ln x \to \dfrac{1}{x}$, $a^x \to (\ln a)\,a^x$, $\sin x \to \cos x$, $\cos x \to -\sin x$, $\tan x \to \sec^2 x$, $\sec x \to \tan x \sec x$, $\operatorname{cosec} x \to -\cot x \operatorname{cosec} x$, $\cot x \to -\operatorname{cosec}^2 x$.
- Choose the rule from the *last* thing done to x: multiplied, divided, a function of a function, or added.
- Inside a product or quotient, each part may need its own chain rule, e.g. $\dfrac{d}{dx}\sin 3x = 3\cos 3x$.
- Simplify by taking out common factors, especially exponentials.
- $e^{f(x)}$, $\ln f(x)$, $\sin f(x)$ and $[f(x)]^n$ all follow the chain rule: keep the inside, multiply by its derivative.



WORKED EXAMPLE

Differentiate $y = e^{-x}\sin 2x$, giving your answer in factorised form.

Last operation: multiplied, so the product rule. — choose the rule

$u = e^{-x}$, $u' = -e^{-x}$; $\quad v = \sin 2x$, $v' = 2\cos 2x$ — chain rule on each part

$$\frac{dy}{dx} = -e^{-x}\sin 2x + 2e^{-x}\cos 2x$$

$$= e^{-x}(2\cos 2x - \sin 2x)$$ — common factor e^{-x}

Tier 1 · products with a chain inside

415. Differentiate:

a) e^x

b) $\ln x$

c) $\sin x$

d) $\cot x$

e) 3^x

416. Which rule comes first?

a) $x^2 \cos x$

b) $\cos\left(x^2\right)$

c) $\dfrac{\cos x}{x^2}$

d) $x^2 + \cos x$

417. Differentiate $y = e^x \cos 2x$, giving your answer in factorised form.

418. Differentiate $y = x^2 e^{3x}$, giving your answer in factorised form.

Tier 2 · products with a negative power

419. Differentiate $y = x^2 e^{-x}$, giving your answer in factorised form.

420. Differentiate $y = xe^{-3x}$, giving your answer in factorised form.

421. Differentiate $y = e^{-x} \cos x$, giving your answer in factorised form.

Tier 3 · quotients

422. Differentiate $y = \dfrac{\ln x}{e^x}$, giving your answer as a single fraction.

423. Differentiate $y = \dfrac{\sin x}{e^{2x}}$.

424. Differentiate $y = \dfrac{x^2}{\ln x}$, giving your answer in factorised form.

Tier 4 · a function of a function

425. Differentiate $y = \cos\left(e^x\right)$.

426. Differentiate $y = e^{\tan x}$.

427. Differentiate $y = \ln\left(1 + e^x\right)$.

Tier 5 · the log of a sum

428. Differentiate $y = \ln(x + \cos x)$.

429. Differentiate $y = \ln\left(e^{x} + e^{-x}\right)$.

430. Differentiate $y = \ln(2x + \sin 2x)$.

Tier 6 · powers of functions

431. Differentiate $y = (e^{x} + 1)^{3}$.

432. Differentiate $y = (\sin x + \cos x)^{2}$.

433. Differentiate $y = \sqrt{1 + \ln x}$.

Tier 7 · quotients with a chain

434. Show that $\dfrac{d}{dx}\left(\dfrac{\sin x}{(1+\cos x)^{2}}\right) = \dfrac{2 - \cos x}{(1+\cos x)^{2}}$.

435. Differentiate $y = \dfrac{x}{\left(x^{2}+1\right)^{2}}$, simplifying fully.

436. Differentiate $y = \dfrac{e^{x}}{\left(e^{x}+1\right)^{3}}$, simplifying fully.

Tier 8 · tangents and normals

437. Find the equation of the tangent to $y = e^{x}\sin x$ at the origin.

438. Find the equation of the tangent to $y = \ln\left(1 + x^{2}\right)$ at the point where $x = 1$.

439. Find the equation of the normal to $y = xe^{2x}$ at the origin.

Conceptual

440. Explain why $y = e^{x}\sin x$ needs the product rule but $y = e^{\sin x}$ needs the chain rule, and differentiate both.

441. Show that $\dfrac{d}{dx}\ln(\sec x) = \tan x$.

442. A student writes $\dfrac{d}{dx}e^{x^{2}} = x^{2}e^{x^{2}-1}$. Explain the error and give the correct derivative.

443. Find the exact gradient of the tangent to $y = x\ln x - x$ at the point where $x = e$.

444. Let $f(x) = \dfrac{\ln x}{x^2}$, for $x > 0$.

(a) Show that $f'(x) = \dfrac{1 - 2\ln x}{x^3}$. (1 mark)

(b) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (2 marks)

Revision

445. Differentiate $y = e^{5x-1}$.

446. Differentiate $y = \ln\left(3x^2\right)$.

447. Differentiate $y = \log_2(x+1)$.

448. Differentiate $y = 7^x$.

449. Differentiate $y = \tan\left(x^2\right)$.

450. Differentiate $y = \sec 3x$.

451. Differentiate $y = \operatorname{cosec} x$.

452. Differentiate $y = \cot 2x$.

453. Differentiate $y = \sin^2 3x$.

454. Differentiate $y = x\cos x$.

455. Find the equation of the tangent to $y = e^{-x}$ at the point where $x = 0$.

456. Find the equation of the normal to $y = \sin x$ at the origin.

457. Show that $\dfrac{d}{dx}\left(\dfrac{e^x - 1}{e^x + 1}\right) = \dfrac{2e^x}{(e^x+1)^2}$.

458. Find the exact value of $\cos\dfrac{3\pi}{4}$.

459. Find the exact value of $\sec\dfrac{\pi}{4}$.

460. Solve $\tan x = -1$ for $0 \le x \le 2\pi$.

461. Solve $\cos 2x = 0$ for $0 \le x \le \pi$.

462. Solve $e^x = 3$, giving the exact answer.

463. Solve $\log_5 x = 2$.

464. Simplify $\ln\left(\dfrac{e^3}{e}\right)$.

465. Differentiate $y = (3x+2)^{-2}$.

466. Differentiate $y = \dfrac{x^2}{1-x}$.

467. Find the equation of the tangent to $y = x^2 + 3x$ at the point where $x = -1$.

468. Solve $2x^2 - 7x + 3 = 0$.

469. Find the coordinates of the vertex of $y = x^2 - 6x + 5$.

470. Solve $|3x - 1| = 5$.

471. State the domain of $f(x) = \sqrt{2x - 6}$.

472. If $P(A) = 0.4$, $P(B) = 0.3$ and $P(A \cap B) = 0.1$, find $P(A \cup B)$.

473. Find the gradient of the line through $(-1, 4)$ and $(3, -4)$.

474. Simplify $\left(x^{-2}\right)^3 \times x^7$.

Solutions

1. a) $f(x) = 9x$, $f'(x) = 9$ b) $f(x) = 2 - x$, $f'(x) = -1$ c) $f(x) = x^4 - 3x$, $f'(x) = 4x^3 - 3$

2. a) $7e^{7x}$ b) $2x\,e^{x^2+1}$ c) $(3x^2 - 6)\,e^{x^3-6x} = 3(x^2 - 2)\,e^{x^3-6x}$

3. a) $-5e^{1-5x}$ b) $\dfrac{1}{2\sqrt{x}}\,e^{\sqrt{x}} = \dfrac{e^{\sqrt{x}}}{2\sqrt{x}}$ c) 0, since e^4 is a constant

4. a) $12e^{3x}$ b) $-10e^{5x+1}$ c) $3e^{6x}$

5. a) $6x\,e^{x^2}$ b) $-(4x - 1)\,e^{2x^2-x} = (1 - 4x)\,e^{2x^2-x}$

6. $\dfrac{d}{dx}\left(\frac{1}{4}e^{8x-2}\right) = \frac{1}{4} \times 8e^{8x-2} = 2e^{8x-2}$

7. a) $4e^{4x} - 4e^{-4x}$ b) $6e^{2x} - 3x^2$

8. $\dfrac{dy}{dx} = -6e^{-3x} - 5e^x + 7$

9. $\dfrac{dy}{dx} = 2x - 4 - e^{2-x}$

10. a) $y = e^{-x}$, so $\dfrac{dy}{dx} = -e^{-x}$ b) $y = 4e^{-2x}$, so $\dfrac{dy}{dx} = -8e^{-2x}$

11. a) $y = e^{4x}$, so $\dfrac{dy}{dx} = 4e^{4x}$ b) $y = e^{6x}$, so $\dfrac{dy}{dx} = 6e^{6x}$

12. a) $y = e^{2x}$, so $\dfrac{dy}{dx} = 2e^{2x}$ b) $y = e^{2x} + 2 + e^{-2x}$, so $\dfrac{dy}{dx} = 2e^{2x} - 2e^{-2x}$

13. $f'(x) = 2e^{2x+1}$, so $f'(0) = 2e$.

14. $f'(x) = 2x\,e^{x^2-4}$, so $f'(2) = 4e^0 = 4$.

15. $\dfrac{dy}{dx} = -3e^{-x} + e^x$. At $x = \ln 3$: $e^{\ln 3} = 3$ and $e^{-\ln 3} = \frac{1}{3}$, so the gradient is $-3 \times \frac{1}{3} + 3 = 2$.

16. $\dfrac{dy}{dx} = 3\left(e^x + 1\right)^2 \times e^x = 3e^x\left(e^x + 1\right)^2$

17. $\dfrac{dy}{dx} = 4\left(2 - e^{3x}\right)^3 \times \left(-3e^{3x}\right) = -12e^{3x}\left(2 - e^{3x}\right)^3$

18. $y = \left(1 + e^{2x}\right)^{\frac{1}{2}}$, so $\dfrac{dy}{dx} = \frac{1}{2}\left(1 + e^{2x}\right)^{-\frac{1}{2}} \times 2e^{2x} = \dfrac{e^{2x}}{\sqrt{1 + e^{2x}}}$

19. a) $y = e^{x^2-2x}$, so $\dfrac{dy}{dx} = (2x - 2)\,e^{x^2-2x} = 2(x - 1)\,e^{x^2-2x}$. b) $e^{x^2-2x} > 0$, so $x = 1$; $y = e^{-1}$. The point is $\left(1,\, e^{-1}\right)$.

20. a) $y = e^{2x^2-8x}$, so $\dfrac{dy}{dx} = (4x - 8)\,e^{2x^2-8x} = 4(x - 2)\,e^{2x^2-8x}$. b) $e^{2x^2-8x} > 0$, so $x = 2$; $y = e^{8-16} = e^{-8}$. The point is $\left(2,\, e^{-8}\right)$.

21. a) $f'(x) = (3x^2 - 3)\,e^{x^3-3x} = 3(x-1)(x+1)\,e^{x^3-3x}$ b) $x = 1$ or $x = -1$. $f(1) = e^{-2}$ and $f(-1) = e^2$, giving $(1, e^{-2})$ and $(-1, e^2)$.

22. The student has used the power rule, which only applies when x is in the base. Here x is in the power, so the power stays the same and is multiplied by its derivative: $\frac{d}{dx}\left(e^{3x}\right) = 3e^{3x}$.

23. Let $u = ax + b$, so $y = e^u$. Then $\frac{du}{dx} = a$ and $\frac{dy}{du} = e^u$, so $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = e^u \times a = ae^{ax+b}$.

24. $\frac{dy}{dx} = 10e^{2x}$. Since $e^{2x} > 0$ for all x, the gradient is always positive, so the graph is always increasing.

25. $\frac{dy}{dx} = (2x + k)\,e^{x^2+kx}$. The exponential is never 0, so $2(3) + k = 0$ and $k = -6$.

26. a) $\frac{dy}{dx} = 10e^{2x}$ b) $10e^{2x} = 2 \times 5e^{2x} = 2y$ c) $\frac{dy}{dx} = 2 \times 10 = 20$

27. $\frac{dy}{dx} = 4(3x-2)^3 \times 3 = 12(3x-2)^3$

28. $y = x^{\frac{1}{2}} + 2x^{-1}$, so $\frac{dy}{dx} = \frac{1}{2\sqrt{x}} - \frac{2}{x^2}$

29. $\frac{dy}{dx} = 2x(x+3)^3 + 3x^2(x+3)^2 = x(x+3)^2(5x+6)$

30. $\frac{dy}{dx} = \frac{2(x-3) - 1(2x+1)}{(x-3)^2} = \frac{-7}{(x-3)^2}$

31. $\frac{dy}{dx} = 3x^2 - 2 = 1$ at $x = 1$, and $y = -1$. So $y + 1 = 1(x-1)$, i.e. $y = x - 2$.

32. $\frac{dy}{dx} = 2x + 4 = 2$ at $x = -1$, so the normal has gradient $-\frac{1}{2}$.

33. $e^{2x-5x-x} = e^{-4x}$

34. $2x = \ln 9$, so $x = \frac{1}{2}\ln 9 = \ln 3$.

35. 5

36. $3 + 0 = 3$

37. $x + 1 = 3^2 = 9$, so $x = 8$.

38. $2\ln x + \ln(x+1)$

39. $\frac{\sqrt{3}}{2}$

40. $x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$

41. 150°

42. $\sin^2\theta$

43. $x > -2$

44. $g(3) = 9$, so $f(9) = 19$.

45. $f(-x) = -x^3 + x = -(x^3 - x) = -f(x)$, so f is odd.

46. $(x-1)(x-5) < 0$, so $1 < x < 5$.

47. $f^{-1}(x) = \dfrac{x+4}{3}$

48. A vertical dilation by a factor of 2 and a translation 1 unit to the right.

49. $0.4 \times 0.5 = 0.2$

50. $\dfrac{6}{36} = \dfrac{1}{6}$

51. $x + 3 = 2x$ gives $x = 3$. $x + 3 = -2x$ gives $x = -1$, but then $2x < 0$, so reject it. $x = 3$.

52. $\dfrac{dy}{dx} = 2x = 1$, so $\tan\theta = 1$ and $\theta = 45°$.

53. a) $e^{x\ln 9}$ b) $e^{5x\ln 2}$

54. a) $x^2e^{2x}(3+2x)$ b) $e^{3x}(1-3x)$

55. a) $(\ln 6)\,6^x$ b) $5(\ln 2)\,2^x$ c) $3^{4x} = e^{4x\ln 3}$, so $\dfrac{dy}{dx} = 4(\ln 3)\,3^{4x}$

56. a) $e^x + xe^x = e^x(1+x)$ b) $2xe^{4x} + 4x^2e^{4x} = 2xe^{4x}(1+2x)$

57. $\dfrac{dy}{dx} = 4x^3e^x + x^4e^x = x^3e^x(4+x)$

58. $\dfrac{dy}{dx} = 3e^{-x} - (3x-1)e^{-x} = e^{-x}(4-3x)$

59. $\dfrac{dy}{dx} = e^{-2x} - 2xe^{-2x} = e^{-2x}(1-2x)$

60. $\dfrac{dy}{dx} = 2xe^{-3x} - 3x^2e^{-3x} = xe^{-3x}(2-3x)$

61. $\dfrac{dy}{dx} = \dfrac{3e^{3x}\cdot x - e^{3x}\cdot 1}{x^2} = \dfrac{e^{3x}(3x-1)}{x^2}$

62. $\dfrac{dy}{dx} = \dfrac{1\cdot e^{2x} - x\cdot 2e^{2x}}{(e^{2x})^2} = \dfrac{e^{2x}(1-2x)}{e^{4x}} = \dfrac{1-2x}{e^{2x}}$

63. $\dfrac{dy}{dx} = \dfrac{e^x(x+1) - e^x\cdot 1}{(x+1)^2} = \dfrac{xe^x}{(x+1)^2}$

64. $\dfrac{dy}{dx} = 2^x + x(\ln 2)\,2^x = 2^x(1 + x\ln 2)$

65. $\dfrac{dy}{dx} = 2x \cdot 5^x + x^2(\ln 5)\,5^x = x \cdot 5^x(2 + x\ln 5)$

66. $\dfrac{dy}{dx} = (\ln 2)\,2^x e^x + 2^x e^x = 2^x e^x(\ln 2 + 1)$

67. $\dfrac{dy}{dx} = \dfrac{e^x(e^x + 2) - e^x \cdot e^x}{(e^x + 2)^2} = \dfrac{2e^x}{(e^x + 2)^2}$

68. $\dfrac{dy}{dx} = \dfrac{e^x(e^x + 1) - (e^x - 1)e^x}{(e^x + 1)^2} = \dfrac{2e^x}{(e^x + 1)^2}$

69. $\dfrac{dy}{dx} = \dfrac{2e^{2x}(1 + e^{2x}) - e^{2x} \cdot 2e^{2x}}{(1 + e^{2x})^2} = \dfrac{2e^{2x}}{(1 + e^{2x})^2}$

70. $\dfrac{dy}{dx} = e^{-x}(1 - x)$. $e^{-x} > 0$, so $x = 1$ and $y = e^{-1}$. The point is $\left(1,\, e^{-1}\right)$.

71. $\dfrac{dy}{dx} = xe^x(x + 2)$. $e^x > 0$, so $x = 0$ or $x = -2$, giving $(0,\, 0)$ and $\left(-2,\, 4e^{-2}\right)$.

72. $\dfrac{dy}{dx} = \dfrac{e^x(x - 1)}{x^2}$, which is 0 when $x = 1$. $y = e$. The point is $(1,\, e)$.

73. $a = e^{\ln a}$, so $a^x = e^{x \ln a}$. Since $\ln a$ is a constant, $\dfrac{d}{dx}\left(e^{x \ln a}\right) = \ln a \cdot e^{x \ln a} = (\ln a)\,a^x$.

74. The student used the power rule, which only applies when x is the base. Here x is the power, so $\dfrac{d}{dx}\left(2^x\right) = (\ln 2)\,2^x$.

75. $\dfrac{d}{dx}\left(xe^x\right) = e^x + xe^x = (x + 1)e^x$. At $x = 0$: $(0 + 1)e^0 = 1$.

76. $\dfrac{dy}{dx} = e^{2x}(1 + 2x)$. At $x = 1$: $e^2(1 + 2) = 3e^2$.

77. a) $f'(x) = e^{-x} - xe^{-x} = (1 - x)e^{-x}$ b) For $x < 1$, $1 - x > 0$ and $e^{-x} > 0$, so $f'(x) > 0$. c) $f'(x) = 0$ when $x = 1$; $f(1) = e^{-1}$. The point is $\left(1,\, e^{-1}\right)$.

78. $4e^{4x-1}$

79. $3\left(e^x + 2\right)^2 \times e^x = 3e^x\left(e^x + 2\right)^2$

80. $f'(x) = 2xe^{x^2-1}$, so $f'(1) = 2e^0 = 2$.

81. $y = e^{-2x}$, so $\dfrac{dy}{dx} = -2e^{-2x}$.

82. -2

83. $27 = 3^3$, so $x + 1 = 3$ and $x = 2$.

84. $x = \ln 5$

85. $3 + 2 = 5$

86. $3\ln x + \ln y$

87. $3(2x-5)^2 \times 2 = 6(2x-5)^2$

88. $y = (3x+1)^{\frac{1}{2}}$, so $\dfrac{dy}{dx} = \frac{1}{2}(3x+1)^{-\frac{1}{2}} \times 3 = \dfrac{3}{2\sqrt{3x+1}}$

89. $\dfrac{dy}{dx} = 2x - 4 = 2$ at $x = 3$, and $y = -3$. So $y + 3 = 2(x-3)$, i.e. $y = 2x - 9$.

90. $\dfrac{dy}{dx} = 3x^2 = 3$, so the normal has gradient $-\frac{1}{3}$.

91. $(x-3)(x+2) = 0$, so $x = 3$ or $x = -2$.

92. $(2x-1)(x+3) = 0$, so $x = \frac{1}{2}$ or $x = -3$.

93. $\frac{1}{2}$

94. $x = \dfrac{\pi}{4}$ or $x = \dfrac{5\pi}{4}$

95. 72°

96. $x > 3$

97. $f(2) = 4$, so $g(4) = 7$.

98. Even: $f(-x) = x^4 + x^2 = f(x)$.

99. $-5 < 2x + 1 < 5$, so $-3 < x < 2$.

100. $0.3 + 0.5 = 0.8$

101. $\left(\dfrac{-2+4}{2}, \dfrac{5-1}{2}\right) = (1, 2)$

102. $\dfrac{(x-3)(x+3)}{x+3} = x - 3$

103. a) e^4 b) $3e^0 = 3$ c) e^3

104. a) $-\dfrac{1}{5e}$ b) $-\dfrac{1}{e^4}$ c) $\dfrac{e}{3}$

105. Point $(2, e^2)$, $m = e^2$. $y - e^2 = e^2(x-2)$, so $y = e^2x - e^2$.

106. Point $(0, 1)$, $m = 5e^0 = 5$. $y - 1 = 5x$, so $y = 5x + 1$.

107. Point $(1, 2e^3)$, $m = 6e^3$. $y - 2e^3 = 6e^3(x-1)$, so $y = 6e^3x - 4e^3$.

108. Point $(0, 1)$. Tangent gradient 3, normal gradient $-\frac{1}{3}$. $y - 1 = -\frac{1}{3}x$, so $x + 3y - 3 = 0$.

109. Point $(1, e)$. Tangent gradient e, normal gradient $-\dfrac{1}{e}$. $e(y-e) = -(x-1)$, so $x + ey - e^2 - 1 = 0$.

110. Point $(0,\,1)$. Tangent gradient -2, normal gradient $\frac{1}{2}$. $y-1=\frac{1}{2}x$, so $x-2y+2=0$.

111. $\dfrac{dy}{dx}=e^{2x}(1+2x)$, so $m=1$ at $x=0$. The tangent is $y=x$.

112. Point $\left(2,\,2e^{-2}\right)$. $\dfrac{dy}{dx}=e^{-x}(1-x)$, so $m=-e^{-2}$. $y-2e^{-2}=-e^{-2}(x-2)$, so $y=-e^{-2}x+4e^{-2}$.

113. Point $(1,\,e)$. $\dfrac{dy}{dx}=xe^{x}(2+x)$, so $m=3e$. $y-e=3e(x-1)$, so $y=3ex-2e$.

114. Point $(-1,\,e)$. $\dfrac{dy}{dx}=2xe^{x^2}=-2e$, so the normal gradient is $\dfrac{1}{2e}$. $2e(y-e)=x+1$, so $x-2ey+2e^2+1=0$.

115. Point $(1,\,1)$. Tangent gradient $2e^{0}=2$, normal gradient $-\frac{1}{2}$. $y-1=-\frac{1}{2}(x-1)$, so $x+2y-3=0$.

116. Point $(2,\,1)$. Tangent gradient $2(2)e^{0}=4$, normal gradient $-\frac{1}{4}$. $y-1=-\frac{1}{4}(x-2)$, so $x+4y-6=0$.

117. Point $\left(a,\,e^{2a}\right)$; $\dfrac{dy}{dx}=2e^{2x}$, so $m=2e^{2a}$. Tangent: $y-e^{2a}=2e^{2a}(x-a)$. At $y=0$: $-e^{2a}=2e^{2a}(x-a)$. Since $e^{2a}\neq 0$, $x-a=-\frac{1}{2}$, so $x=a-\frac{1}{2}$.

118. Point $(a,\,e^{-a})$; $\dfrac{dy}{dx}=-e^{-x}$, so $m=-e^{-a}$. Tangent: $y-e^{-a}=-e^{-a}(x-a)$. At $y=0$: $-e^{-a}=-e^{-a}(x-a)$. Since $e^{-a}\neq 0$, $x-a=1$, so $x=a+1$.

119. Tangent: $y-e^{a}=e^{a}(x-a)$. At $x=0$: $y=e^{a}-ae^{a}=(1-a)e^{a}$.

120. Tangent: $y-e^{2a}=2e^{2a}(x-a)$. Through $(0,\,0)$: $-e^{2a}=-2ae^{2a}$, and $e^{2a}>0$, so $a=\frac{1}{2}$. The point is $\left(\frac{1}{2},\,e\right)$ with $m=2e$, so the tangent is $y=2ex$.

121. $e^{x}=3$, so $x=\ln 3$ and $y=3$. The point is $(\ln 3,\,3)$.

122. The tangent gradient is 6: $3e^{3x}=6$, so $e^{3x}=2$ and $x=\frac{1}{3}\ln 2$. Then $y=2$. The point is $\left(\frac{1}{3}\ln 2,\,2\right)$.

123. A horizontal tangent has gradient 0. The gradient is $\dfrac{dy}{dx}=e^{x}$, which is greater than 0 for every x, so it is never 0.

124. For $y=e^{x}$: $m_1=e^{0}=1$. For $y=e^{-x}$: $m_2=-e^{0}=-1$. $m_1m_2=-1$, so the tangents are perpendicular.

125. Point $(0,\,1)$; tangent gradient 1, normal gradient -1. Normal: $y=-x+1$, which crosses at $(1,\,0)$.

126. The y-value must come from the original function, $y=e^{2}$, not from the derivative $2e^{2}$. The correct point is $\left(1,\,e^{2}\right)$.

127. a) Point $(0,\,1)$, $f'(x)=2e^{2x}$ so $m=2$. Tangent: $y=2x+1$. b) $2x+1=0$, so $x=-\frac{1}{2}$: the point $\left(-\frac{1}{2},\,0\right)$.

128. $e^{3x}+3xe^{3x}=e^{3x}(1+3x)$

129. $(\ln 4)\,4^{x}$

130. $\dfrac{xe^x - e^x}{x^2} = \dfrac{e^x(x-1)}{x^2}$

131. $3x^2e^{x^3}$

132. $2\left(1-e^x\right) \times \left(-e^x\right) = -2e^x\left(1-e^x\right)$

133. $2x = 5$, so $x = \frac{5}{2}$.

134. $x = \log_2 10 = \dfrac{\ln 10}{\ln 2}$

135. 7

136. 4

137. $y = \frac{2}{5}x + \frac{1}{5}$, so the gradient is $\frac{2}{5}$.

138. Gradient $-\frac{1}{2}$: $y - 3 = -\frac{1}{2}(x+1)$, so $x + 2y - 5 = 0$.

139. $\sqrt{3^2+4^2} = 5$

140. $4\left(x^2+1\right)^3 \times 2x = 8x\left(x^2+1\right)^3$

141. $\dfrac{(x-1)-x}{(x-1)^2} = -\dfrac{1}{(x-1)^2}$

142. $x(x-4) = 0$, so $x = 0$ or $x = 4$.

143. $x = \dfrac{-2 \pm \sqrt{4+8}}{2} = -1 \pm \sqrt{3}$

144. $\dfrac{\sqrt{2}}{2}$

145. $\cos x = \frac{1}{2}$, so $x = \dfrac{\pi}{3}$ or $x = \dfrac{5\pi}{3}$.

146. $y > 2$

147. $y = 3$

148. $f^{-1}(x) = \ln x$

149. $x - 3 = 5$ or $x - 3 = -5$, so $x = 8$ or $x = -2$.

150. $1 - 0.6 = 0.4$

151. $x^3 + 6x^2 + 12x + 8$

152. $\dfrac{4x^6}{4x^2} = x^4$

153. a) $f(x) = 9x$, $f'(x) = 9$ b) $f(x) = x^2 - 7$, $f'(x) = 2x$ c) $f(x) = 5 - 2x$, $f'(x) = -2$

154. a) $3\ln x + \ln(x+2)$ b) $\ln(x-1) - \ln(x+5)$ c) $2\ln(4x+3)$

155. a) Yes (product) b) No (a sum) c) Yes (power)

156. a) $\dfrac{2x}{x^2+1}$ b) $\dfrac{3x^2-4}{x^3-4x}$

157. $\dfrac{dy}{dx} = \dfrac{4x-5}{2x^2-5x+3}$

158. $\dfrac{dy}{dx} = \dfrac{6}{6x} = \dfrac{1}{x}$

159. $\dfrac{dy}{dx} = \dfrac{3}{3x+2}$

160. $\dfrac{dy}{dx} = \dfrac{-4}{1-4x} = \dfrac{4}{4x-1}$

161. $\dfrac{dy}{dx} = 5 \times \dfrac{2}{2x+3} = \dfrac{10}{2x+3}$

162. $\dfrac{dy}{dx} = 3x^2 - \dfrac{2}{x}$

163. $\dfrac{dy}{dx} = \dfrac{2x}{x^2+1} + 2e^{2x}$

164. $y = 3\ln(2x+1)$, so $\dfrac{dy}{dx} = \dfrac{6}{2x+1}$.

165. $y = \frac{1}{2}\ln(4x-1)$, so $\dfrac{dy}{dx} = \frac{1}{2} \times \dfrac{4}{4x-1} = \dfrac{2}{4x-1}$.

166. $y = 5\ln x$, so $\dfrac{dy}{dx} = \dfrac{5}{x}$.

167. $y = \ln x + \ln(x+3)$, so $\dfrac{dy}{dx} = \dfrac{1}{x} + \dfrac{1}{x+3} = \dfrac{2x+3}{x(x+3)}$.

168. $\dfrac{dy}{dx} = \dfrac{1}{x+2} - \dfrac{1}{x-1} = \dfrac{(x-1)-(x+2)}{(x+2)(x-1)} = \dfrac{-3}{(x+2)(x-1)}$

169. $\dfrac{dy}{dx} = \dfrac{3}{3x-1} - \dfrac{1}{x+1} = \dfrac{3(x+1)-(3x-1)}{(3x-1)(x+1)} = \dfrac{4}{(3x-1)(x+1)}$

170. $y = 3\ln x + \frac{1}{2}\ln(x+1)$, so $\dfrac{dy}{dx} = \dfrac{3}{x} + \dfrac{1}{2(x+1)} = \dfrac{7x+6}{2x(x+1)}$.

171. $y = 2\ln(x+1) - \ln(x-2)$, so $\dfrac{dy}{dx} = \dfrac{2}{x+1} - \dfrac{1}{x-2} = \dfrac{x-5}{(x+1)(x-2)}$.

172. $y = \frac{1}{2}\ln(x+1) - \frac{1}{2}\ln(x-1)$, so $\dfrac{dy}{dx} = \frac{1}{2}\left(\dfrac{1}{x+1} - \dfrac{1}{x-1}\right) = \frac{1}{2} \times \dfrac{-2}{x^2-1} = -\dfrac{1}{x^2-1}$.

173. Differentiate both sides with the chain rule: $e^{\ln x} \times \dfrac{d}{dx}(\ln x) = 1$. Since $e^{\ln x} = x$, this gives $x\dfrac{d}{dx}(\ln x) = 1$, so $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$.

174. $\dfrac{d}{dx}\ln(5x) = \dfrac{5}{5x} = \dfrac{1}{x}$. By the log laws $\ln(5x) = \ln 5 + \ln x$, and $\ln 5$ is a constant with derivative 0.

175. The student forgot to multiply by the derivative of the inside. Correct: $\dfrac{2x}{x^2+1}$.

176. $\dfrac{dy}{dx} = \dfrac{2x}{x^2+3} = \dfrac{2}{4} = \dfrac{1}{2}$ at $x = 1$.

177. a) $f'(x) = \dfrac{2x-2}{x^2-2x+5}$ b) $x^2 - 2x + 5 = (x-1)^2 + 4 > 0$ for all x, so the log is always defined. c) $f'(x) = 0$ when $x = 1$; $f(1) = \ln 4$. The point is $(1,\, \ln 4)$.

178. $3e^{3x} - e^{-x}$

179. $2xe^x + x^2e^x = xe^x(2+x)$

180. $3(\ln 5)\,5^x$

181. Point $(0,\, 1)$, $m = 1$: $y = x + 1$.

182. Point $(0,\, 1)$, tangent gradient 1, normal gradient -1: $y = -x + 1$.

183. $\dfrac{dA}{dt} = 10e^{0.1t}$, so 10 at $t = 0$.

184. $e^{\ln 9} = 9$

185. $x = e^2$

186. $x + 1 = 1$, so $x = 0$.

187. $\ln 9 + \ln 4 = \ln 36$

188. $x = 2^5 = 32$

189. $\dfrac{\ln 20}{\ln 5}$

190. $5\left(3x^2-1\right)^4 \times 6x = 30x\left(3x^2-1\right)^4$

191. $y = x^{\frac{3}{2}}$, so $\dfrac{dy}{dx} = \frac{3}{2}x^{\frac{1}{2}} = \frac{3}{2}\sqrt{x}$.

192. $2x - 6 = 4$, so $x = 5$.

193. $(x-3)(x-4) = 0$, so $x = 3$ or $x = 4$.

194. $\Delta = 1 - 12 = -11 < 0$, so no real roots.

195. $\sqrt{3}$

196. $x = \dfrac{7\pi}{6}$ or $x = \dfrac{11\pi}{6}$

197. $x \leq 4$

198. All real y.

199. $3x > 9$, so $x > 3$.

200. $2x - 1 = 5$, so $f^{-1}(5) = 3$.

201. $P(A \cap B) = 0.2$, so $P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7$.

202. $\dfrac{x(x-4)}{(x-4)(x+4)} = \dfrac{x}{x+4}$

203. a) $\dfrac{\ln x}{\ln 6}$ b) $\dfrac{\ln\left(x^2-1\right)}{\ln 3}$

204. a) $\dfrac{1}{x\ln 4}$ b) $\dfrac{3}{x\ln 5}$ c) $\dfrac{3}{(3x-1)\ln 2}$

205. a) Log law b) Chain rule c) Chain rule d) Log law

206. $\dfrac{dy}{dx} = 1 \times \ln x + x \times \dfrac{1}{x} = \ln x + 1$

207. $\dfrac{dy}{dx} = 5x^4 \ln x + x^4 = x^4(5\ln x + 1)$

208. $\dfrac{dy}{dx} = e^x \ln x + \dfrac{e^x}{x} = e^x\left(\ln x + \dfrac{1}{x}\right)$

209. $\dfrac{dy}{dx} = 2x\ln(3x) + x^2 \times \dfrac{1}{x} = x\big(2\ln(3x) + 1\big)$

210. $\dfrac{dy}{dx} = \dfrac{\ln x}{2\sqrt{x}} + \dfrac{\sqrt{x}}{x} = \dfrac{\ln x}{2\sqrt{x}} + \dfrac{2}{2\sqrt{x}} = \dfrac{\ln x + 2}{2\sqrt{x}}$

211. $\dfrac{dy}{dx} = \dfrac{\frac{1}{x}\cdot x^2 - \ln x \cdot 2x}{x^4} = \dfrac{x(1-2\ln x)}{x^4} = \dfrac{1-2\ln x}{x^3}$

212. $\dfrac{dy}{dx} = \dfrac{1\cdot \ln x - x\cdot \frac{1}{x}}{(\ln x)^2} = \dfrac{\ln x - 1}{(\ln x)^2}$

213. $\dfrac{dy}{dx} = \dfrac{\frac{1}{x}(x+1) - \ln x}{(x+1)^2} = \dfrac{x+1-x\ln x}{x(x+1)^2}$

214. $\dfrac{dy}{dx} = 2\ln x \times \dfrac{1}{x} = \dfrac{2\ln x}{x}$

215. $\dfrac{dy}{dx} = 4(\ln x)^3 \times \dfrac{1}{x} = \dfrac{4(\ln x)^3}{x}$

216. $y = (\ln x)^{\frac{1}{2}}$, so $\dfrac{dy}{dx} = \frac{1}{2}(\ln x)^{-\frac{1}{2}} \times \dfrac{1}{x} = \dfrac{1}{2x\sqrt{\ln x}}$.

217. Point $(1, 0)$, $m = 1$. The tangent is $y = x - 1$.

218. Point $\left(\frac{1}{3}, \ln 1\right) = \left(\frac{1}{3}, 0\right)$. $\dfrac{dy}{dx} = \dfrac{1}{x}$, so $m = 3$. $y = 3\left(x - \frac{1}{3}\right) = 3x - 1$.

219. Point $(e, 2)$. $\dfrac{dy}{dx} = \dfrac{2}{x}$, so $m = \dfrac{2}{e}$. $y - 2 = \dfrac{2}{e}(x - e)$, so $y = \dfrac{2x}{e}$.

220. Point $(e, 1)$, tangent gradient $\dfrac{1}{e}$, normal gradient $-e$. $y - 1 = -e(x - e)$, so $y = -ex + e^2 + 1$.

221. Point $(1, 0)$. $\dfrac{dy}{dx} = x(2\ln x + 1)$, so $m = 1$. The tangent is $y = x - 1$.

222. Point $(1, 0)$. $\dfrac{dy}{dx} = \ln x + 1 = 1$, so the normal gradient is -1. The normal is $y = -x + 1$.

223. By the change of base law, $\log_a x = \dfrac{\ln x}{\ln a} = \dfrac{1}{\ln a} \cdot \ln x$. Since $\dfrac{1}{\ln a}$ is a constant, $\dfrac{d}{dx}(\log_a x) = \dfrac{1}{\ln a} \cdot \dfrac{1}{x} = \dfrac{1}{x \ln a}$.

224. $\ln(x^2) = 2\ln x$, with derivative $\dfrac{2}{x}$. $(\ln x)^2$ needs the chain rule: $\dfrac{2\ln x}{x}$. In the first the power is inside the log; in the second it is on the whole log.

225. Point $(e, 1)$, $m = \dfrac{1}{e}$. $y - 1 = \dfrac{1}{e}(x - e)$ gives $y = \dfrac{x}{e}$, which passes through $(0, 0)$.

226. $\dfrac{dy}{dx} = \ln x + 1 = 0$, so $\ln x = -1$ and $x = e^{-1}$. $y = e^{-1}\ln e^{-1} = -e^{-1}$. The point is $(e^{-1}, -e^{-1})$.

227. a) $f'(x) = \dfrac{\frac{1}{x} \cdot x - \ln x \cdot 1}{x^2} = \dfrac{1 - \ln x}{x^2}$ b) $1 - \ln x = 0$, so $x = e$; $f(e) = \dfrac{1}{e}$. The point is $\left(e, \dfrac{1}{e}\right)$.

228. $\dfrac{2x}{x^2 + 7}$

229. $\dfrac{1}{x+1} - \dfrac{1}{x-1} = \dfrac{-2}{x^2 - 1}$

230. $e^{-x} - xe^{-x} = e^{-x}(1 - x)$

231. $(\ln 10)\, 10^x$

232. $\dfrac{1}{2\sqrt{x}} e^{\sqrt{x}} = \dfrac{e^{\sqrt{x}}}{2\sqrt{x}}$

233. Point $(0, 1)$, $m = 2$: $y = 2x + 1$.

234. $\dfrac{d}{dx}\left(e^{2x} - e^{-2x}\right) = 2e^{2x} - \left(-2e^{-2x}\right) = 2e^{2x} + 2e^{-2x} = 2\left(e^{2x} + e^{-2x}\right)$

235. $(e^x - 1)(e^x - 2) = 0$, so $e^x = 1$ or $e^x = 2$: $x = 0$ or $x = \ln 2$.

236. $2x - 1 = 1$, so $x = 1$.

237. $x + x = 2x$

238. $\frac{1}{2}$, since $4^{\frac{1}{2}} = 2$.

239. $\ln\left(\dfrac{18 \times 5}{2}\right) = \ln 45$

240. $\dfrac{4(2x-3)-2(4x+1)}{(2x-3)^2} = \dfrac{-14}{(2x-3)^2}$

241. $y = 6x^{-\frac{1}{2}}$, so $\dfrac{dy}{dx} = -3x^{-\frac{3}{2}}$.

242. Tangent gradient 2, normal gradient $-\frac{1}{2}$: $y-1 = -\frac{1}{2}(x-1)$, so $x+2y-3=0$.

243. $x(x-2)(x+2) = 0$, so $x = 0, 2$ or -2.

244. $\Delta = k^2 - 36 = 0$, so $k = \pm 6$.

245. $-\dfrac{\sqrt{3}}{2}$

246. $\sin x = \pm\dfrac{1}{\sqrt{2}}$, so $x = \dfrac{\pi}{4}, \dfrac{3\pi}{4}, \dfrac{5\pi}{4}, \dfrac{7\pi}{4}$.

247. $\dfrac{2\pi}{3}$

248. All real x except $x = 2$.

249. $x+1 \geq 3$ or $x+1 \leq -3$, so $x \geq 2$ or $x \leq -4$.

250. $g(4) = 25$, so $f(25) = 5$.

251. $0.3 \times 0.5 = 0.15$

252. $(x-2)\left(x^2+2x+4\right)$

253. a) $f(x) = 9x$, $f'(x) = 9$ b) $f(x) = x^2 - 1$, $f'(x) = 2x$ c) $f(x) = 5 - 2x$, $f'(x) = -2$

254. a) $\cos x - \sin x$ b) $4\sec^2 x$ c) $2x + 3\sin x$

255. $\dfrac{dy}{dx} = (2x+3)\cos\left(x^2+3x\right)$

256. $\dfrac{dy}{dx} = 3x^2\cos\left(x^3\right)$

257. $\dfrac{dy}{dx} = 5\cos(5x-2)$

258. $\dfrac{dy}{dx} = -2x\sin\left(x^2+1\right)$

259. $\dfrac{dy}{dx} = -4\sin 4x$

260. $\dfrac{dy}{dx} = -(-3)\sin(2-3x) = 3\sin(2-3x)$

261. $\dfrac{dy}{dx} = 6\sec^2 6x$

262. $\dfrac{dy}{dx} = 2x\sec^2\left(x^2\right)$

263. $\dfrac{dy}{dx} = -\sec^2(1-x)$

264. $\dfrac{dy}{dx} = 15\cos 3x - 8\sin 4x$

265. $\dfrac{dy}{dx} = 2 \times \frac{1}{2}\sec^2\dfrac{x}{2} = \sec^2\dfrac{x}{2}$

266. $\dfrac{dy}{dx} = -\pi\sin(\pi x) - \pi\cos(\pi x)$

267. $\dfrac{dy}{dx} = -2\sin 2x$. At $x = \dfrac{\pi}{12}$: $-2\sin\dfrac{\pi}{6} = -1$.

268. $\dfrac{dy}{dx} = \sec^2 x = \dfrac{1}{\cos^2 x}$. At $x = \dfrac{\pi}{4}$: $\dfrac{1}{\left(\frac{1}{\sqrt{2}}\right)^2} = 2$.

269. $\dfrac{dy}{dx} = 3\cos 3x$. At $x = \dfrac{\pi}{9}$: $3\cos\dfrac{\pi}{3} = \dfrac{3}{2}$.

270. $\dfrac{dy}{dx} = \cos x + \sin x = 0$, so $\tan x = -1$: $x = \dfrac{3\pi}{4}$ or $\dfrac{7\pi}{4}$.

271. $\dfrac{dy}{dx} = 1 - 2\sin x = 0$, so $\sin x = \frac{1}{2}$: $x = \dfrac{\pi}{6}$ or $\dfrac{5\pi}{6}$.

272. $\dfrac{dy}{dx} = 2\cos 2x = 0$, so $2x = \dfrac{\pi}{2}$ or $\dfrac{3\pi}{2}$: $x = \dfrac{\pi}{4}$ or $\dfrac{3\pi}{4}$.

273. $\tan x = \dfrac{\sin x}{\cos x}$, so $\dfrac{d}{dx}(\tan x) = \dfrac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} = \dfrac{\cos^2 x + \sin^2 x}{\cos^2 x} = \dfrac{1}{\cos^2 x} = \sec^2 x$.

274. At $x = 0$ the curve rises most steeply, with gradient 1; at $x = \frac{\pi}{2}$ it reaches a maximum, so the tangent is horizontal; at $x = \pi$ it falls most steeply, with gradient -1. These are the values of $\cos x$ at 0, $\frac{\pi}{2}$ and π.

275. The student forgot to multiply by the derivative of the angle, 3. Correct: $-3\sin 3x$.

276. $\dfrac{dy}{dx} = \sec^2 x = \dfrac{1}{\cos^2 x}$. Since $0 < \cos^2 x \leq 1$ wherever $\tan x$ is defined, the gradient is always at least 1, so it is never 0.

277. a) $f'(x) = \cos x - \sqrt{3}\sin x$ b) $\cos x = \sqrt{3}\sin x$, so $\tan x = \dfrac{1}{\sqrt{3}}$: $x = \dfrac{\pi}{6}$ or $\dfrac{7\pi}{6}$.

278. $\dfrac{5}{5x-2}$

279. $\ln x + 1$

280. $\dfrac{1}{x\ln 3}$

281. $\dfrac{2\ln x}{x}$

282. $-3e^{-3x}$

283. $3x^2e^{2x} + 2x^3e^{2x} = x^2e^{2x}(3 + 2x)$

284. Point $(e^2, 2)$, $m = \dfrac{1}{e^2}$. $y - 2 = \dfrac{1}{e^2}(x - e^2)$, so $y = \dfrac{x}{e^2} + 1$.

285. $\dfrac{5\pi}{4}$

286. $210°$

287. $-\dfrac{\sqrt{2}}{2}$

288. $-\sqrt{3}$

289. $x = \dfrac{2\pi}{3}$ or $x = \dfrac{4\pi}{3}$

290. $x = \dfrac{\pi}{3}$ or $x = \dfrac{4\pi}{3}$

291. $\dfrac{\sin^2 x}{\sin x} = \sin x$

292. Amplitude 3, period π.

293. $\ell = r\theta = 6 \times \dfrac{\pi}{3} = 2\pi$ cm

294. $A = \frac{1}{2}r^2\theta = \frac{1}{2} \times 16 \times \dfrac{\pi}{2} = 4\pi$ cm^2

295. $32 = 2^5$, so $x + 1 = 5$ and $x = 4$.

296. $\ln(2x) = \ln 10$, so $2x = 10$ and $x = 5$.

297. $4\left(x^2 - 3\right)^3 \times 2x = 8x\left(x^2 - 3\right)^3$

298. Tangent gradient 3, normal gradient $-\frac{1}{3}$: $y - 1 = -\frac{1}{3}(x - 1)$, so $x + 3y - 4 = 0$.

299. $(x + 5)(x - 1) < 0$, so $-5 < x < 1$.

300. Odd: $\sin(-x) = -\sin x$.

301. $\dfrac{6}{36} = \dfrac{1}{6}$

302. $2x^2 + 5x - 3$

303. a) $(\cos x)^3$ b) $(\sin 4x)^2$

304. $\dfrac{dy}{dx} = \sin x + x\cos x$

305. $\dfrac{dy}{dx} = 2x\cos x - x^2\sin x = x(2\cos x - x\sin x)$

306. $\dfrac{dy}{dx} = 3x^2 \tan x + x^3 \sec^2 x = x^2\left(3\tan x + x\sec^2 x\right)$

307. $\dfrac{dy}{dx} = \dfrac{\sin x}{2\sqrt{x}} + \sqrt{x}\cos x$

308. $\dfrac{dy}{dx} = \sin 3x + 3x\cos 3x$

309. $\dfrac{dy}{dx} = 2x\cos 2x - 2x^2 \sin 2x = 2x(\cos 2x - x\sin 2x)$

310. $\dfrac{dy}{dx} = \cos x \cdot \cos x + \sin x \cdot (-\sin x) = \cos^2 x - \sin^2 x$

311. $\dfrac{dy}{dx} = 2\cos x \times (-\sin x) = -2\sin x \cos x$

312. $\dfrac{dy}{dx} = 3\sin^2 x \cos x$

313. $\dfrac{dy}{dx} = 2\tan x \sec^2 x$

314. $\dfrac{dy}{dx} = \dfrac{-x\sin x - \cos x}{x^2}$

315. $\dfrac{dy}{dx} = \dfrac{\sin x - x\cos x}{\sin^2 x}$

316. $\dfrac{\cos x(1+\cos x) - \sin x(-\sin x)}{(1+\cos x)^2} = \dfrac{\cos x + \cos^2 x + \sin^2 x}{(1+\cos x)^2} = \dfrac{1+\cos x}{(1+\cos x)^2} = \dfrac{1}{1+\cos x}$

317. $\dfrac{dy}{dx} = e^x \sin x + e^x \cos x = e^x(\sin x + \cos x)$

318. $\dfrac{dy}{dx} = \dfrac{\cos x}{\sin x}$

319. $\dfrac{dy}{dx} = -\sin x\, e^{\cos x}$

320. Point $(\pi, -\pi)$. $\dfrac{dy}{dx} = \cos x - x\sin x = -1$. $y + \pi = -(x - \pi)$, so $y = -x$.

321. Point $\left(\dfrac{\pi}{4}, \dfrac{1}{2}\right)$. $\dfrac{dy}{dx} = 2\sin x \cos x = 2 \times \dfrac{1}{\sqrt{2}} \times \dfrac{1}{\sqrt{2}} = 1$. $y = x - \dfrac{\pi}{4} + \dfrac{1}{2}$.

322. Point $(0, 1)$. $\dfrac{dy}{dx} = e^x(\cos x - \sin x) = 1$, so the normal gradient is -1: $y = -x + 1$.

323. Quotient rule: $u = \cos x$, $u' = -\sin x$; $v = 1 - \sin x$, $v' = -\cos x$.
$\dfrac{-\sin x(1-\sin x) + \cos^2 x}{(1-\sin x)^2} = \dfrac{-\sin x + \sin^2 x + \cos^2 x}{(1-\sin x)^2} = \dfrac{1-\sin x}{(1-\sin x)^2} = \dfrac{1}{1-\sin x}$

324. $\dfrac{d}{dx}(x\sin x) = \sin x + x\cos x$ (product rule) and $\dfrac{d}{dx}(\cos x) = -\sin x$. Sum: $\sin x + x\cos x - \sin x = x\cos x$.

325. Product rule: $u = e^{-x}$, $u' = -e^{-x}$; $v = \sin x + \cos x$, $v' = \cos x - \sin x$.
$-e^{-x}(\sin x + \cos x) + e^{-x}(\cos x - \sin x) = e^{-x}(-2\sin x) = -2e^{-x}\sin x$

326. $2\sin x\cos x + 2\cos x(-\sin x) = 0$. This makes sense because $\sin^2 x + \cos^2 x = 1$, a constant.

327. $\dfrac{d}{dx}\ln(\sin x) = \dfrac{\cos x}{\sin x} = \cot x$

328. The derivative of a product is not the product of the derivatives. By the product rule, $\dfrac{d}{dx}(x\sin x) = \sin x + x\cos x$.

329. $\dfrac{dy}{dx} = 2x\sin x + x^2\cos x$. At $x = \pi$: $0 + \pi^2(-1) = -\pi^2$.

330. a) $f'(x) = -e^{-x}\sin x + e^{-x}\cos x = e^{-x}(\cos x - \sin x)$ b) $e^{-x} > 0$, so $\cos x = \sin x$, $\tan x = 1$: $x = \dfrac{\pi}{4}$ or $\dfrac{5\pi}{4}$.

331. $2x\cos\left(x^2\right)$

332. $5\sec^2 5x$

333. $-3\sin(3x+1)$

334. $-\sin\dfrac{\pi}{6} = -\dfrac{1}{2}$

335. $2xe^{x^2}$

336. $\dfrac{3x^2}{x^3+1}$

337. $3(\ln 2)\,2^{3x}$

338. $\dfrac{1}{x\ln 10}$

339. $e^x(1+x)$

340. $\dfrac{1-\ln x}{x^2}$

341. Point $(1, e)$, $m = e$: $y - e = e(x-1)$, so $y = ex$.

342. Point $(1, 0)$, tangent gradient 1, normal gradient -1: $y = -x + 1$.

343. $-\dfrac{\sqrt{3}}{2}$

344. $2x = 0,\ \pi,\ 2\pi$, so $x = 0,\ \dfrac{\pi}{2},\ \pi$.

345. $\cos x = -\dfrac{\sqrt{3}}{2}$, so $x = \dfrac{5\pi}{6}$ or $\dfrac{7\pi}{6}$.

346. $\sec^2 x$

347. $-1 \leq y \leq 3$

348. $\dfrac{\pi}{2}$

349. $e^x = e^{-1}$, so $x = -1$.

350. $2 + \ln x - \ln x = 2$

351. $\log_2 16 = 4$

352. $5(1-2x)^4 \times (-2) = -10(1-2x)^4$

353. $\dfrac{2x(x+1) - x^2}{(x+1)^2} = \dfrac{x^2 + 2x}{(x+1)^2}$

354. $x = \dfrac{2 \pm \sqrt{8}}{2} = 1 \pm \sqrt{2}$

355. The line has gradient $-\frac{3}{4}$, so the perpendicular gradient is $\frac{4}{3}$.

356. $(a+1)^2 - 1 = a^2 + 2a$

357. $1 - \left(\frac{1}{2}\right)^3 = \frac{7}{8}$

358. $5\sqrt{2} - 3\sqrt{2} = 2\sqrt{2}$

359. a) $(\sin 5x)^{-1}$ b) $\left(\cos\left(x^3\right)\right)^{-1}$

360. a) $\dfrac{2}{\sqrt{3}}$ b) $\dfrac{2}{\sqrt{3}}$ c) 1 d) 1

361. $\dfrac{d}{dx}(\cos 3x)^{-1} = -(\cos 3x)^{-2} \times (-3\sin 3x) = \dfrac{3\sin 3x}{\cos^2 3x} = 3\tan 3x \sec 3x$

362. $y = (\cos x)^{-2}$, so $\dfrac{dy}{dx} = -2(\cos x)^{-3}(-\sin x) = \dfrac{2\sin x}{\cos^3 x} = 2\tan x \sec^2 x.$

363. $\dfrac{dy}{dx} = 5\tan x \sec x$

364. $\dfrac{dy}{dx} = -\cot x \operatorname{cosec} x + 2x$

365. $\dfrac{dy}{dx} = -2\cot x \operatorname{cosec} x$

366. $\dfrac{d}{dx}(\sin 2x)^{-1} = -(\sin 2x)^{-2} \times 2\cos 2x = -2 \cdot \dfrac{\cos 2x}{\sin 2x} \cdot \dfrac{1}{\sin 2x} = -2\cot 2x \operatorname{cosec} 2x$

367. $\dfrac{dy}{dx} = -\operatorname{cosec}^2 x - 1$

368. $\dfrac{dy}{dx} = -4\operatorname{cosec}^2 x$

369. $\dfrac{d}{dx}(\tan x)^{-1} = -(\tan x)^{-2}\sec^2 x = -\dfrac{\cos^2 x}{\sin^2 x}\cdot\dfrac{1}{\cos^2 x} = -\dfrac{1}{\sin^2 x} = -\operatorname{cosec}^2 x$

370. $\dfrac{dy}{dx} = \tan x \sec x - \cot x \operatorname{cosec} x$

371. $\dfrac{dy}{dx} = -2\operatorname{cosec}^2 x - 3\tan x \sec x$

372. $\dfrac{dy}{dx} = \sec^2 x - \operatorname{cosec}^2 x$

373. $\dfrac{dy}{dx} = 2x\tan\left(x^2\right)\sec\left(x^2\right)$

374. $\dfrac{dy}{dx} = -3\operatorname{cosec}^2 3x$

375. $\dfrac{dy}{dx} = -2\cot(2x-1)\operatorname{cosec}(2x-1)$

376. $\dfrac{dy}{dx} = \operatorname{cosec} x - x\cot x \operatorname{cosec} x = \operatorname{cosec} x(1 - x\cot x)$

377. $\dfrac{dy}{dx} = 2x\cot x - x^2\operatorname{cosec}^2 x = x\left(2\cot x - x\operatorname{cosec}^2 x\right)$

378. $\dfrac{dy}{dx} = e^x\sec x + e^x\tan x\sec x = e^x\sec x(1+\tan x)$

379. Point $\left(\dfrac{\pi}{3}, 2\right)$. $m = \tan\dfrac{\pi}{3}\sec\dfrac{\pi}{3} = \sqrt{3}\times 2 = 2\sqrt{3}$. $y - 2 = 2\sqrt{3}\left(x - \dfrac{\pi}{3}\right)$, so $y = 2\sqrt{3}x - \dfrac{2\sqrt{3}\pi}{3} + 2$.

380. Tangent gradient $-\cot\dfrac{\pi}{4}\operatorname{cosec}\dfrac{\pi}{4} = -\sqrt{2}$, so the normal gradient is $\dfrac{1}{\sqrt{2}}$.

381. Point $\left(\dfrac{\pi}{2}, 0\right)$. $m = -\operatorname{cosec}^2\dfrac{\pi}{2} = -1$. The tangent is $y = -x + \dfrac{\pi}{2}$.

382. $\sec x = (\cos x)^{-1}$, so $\dfrac{d}{dx}(\sec x) = -(\cos x)^{-2}(-\sin x) = \dfrac{\sin x}{\cos x}\cdot\dfrac{1}{\cos x} = \tan x\sec x$.

383. $\cot x = \dfrac{\cos x}{\sin x}$, so $\dfrac{d}{dx}(\cot x) = \dfrac{-\sin^2 x - \cos^2 x}{\sin^2 x} = -\dfrac{1}{\sin^2 x} = -\operatorname{cosec}^2 x$.

384. $\dfrac{dy}{dx} = -\operatorname{cosec}^2 x = -\dfrac{1}{\sin^2 x}$. Since $\sin^2 x > 0$ wherever $\cot x$ is defined, the gradient is always negative.

385. The derivative of a reciprocal is not the reciprocal of the derivative. Using the chain rule on $(\cos x)^{-1}$ gives $\dfrac{\sin x}{\cos^2 x} = \tan x\sec x$.

386. a) $f'(x) = -(\cos x)^{-2}(-\sin x) = \dfrac{\sin x}{\cos^2 x}$ b) $\sin x = 0$ gives $x = 0$; the point is $(0, 1)$. c) For $0 < x < \dfrac{\pi}{2}$, $\sin x > 0$ and $\cos^2 x > 0$.

387. $\sin x + x\cos x$

388. $-2\sin x\cos x$

389. $3\sec^2 3x$

390. $\cos x\, e^{\sin x}$

391. $\dfrac{x\cos x - \sin x}{x^2}$

392. $\dfrac{2x}{x^2+1}$

393. $(\ln 5)\, 5^x$

394. $xe^{-x}(2-x)$

395. Point $(\pi, 0)$, $m = \cos\pi = -1$: $y = -x + \pi$.

396. $\sec^2 \dfrac{\pi}{3} = 2^2 = 4$

397. $\tan x = 1$, so $x = \dfrac{\pi}{4}$ or $\dfrac{5\pi}{4}$.

398. $-\dfrac{\sqrt{3}}{2}$

399. $\dfrac{5\pi}{3}$

400. 1

401. $\operatorname{cosec}^2 x$

402. $\sin x = \frac{1}{2}$, so $x = \dfrac{\pi}{6}$ or $\dfrac{5\pi}{6}$.

403. $x - 2 = 5$, so $x = 7$.

404. $e^{2x} = 4$, so $2x = \ln 4$ and $x = \ln 2$.

405. $\log_3 1 = 0$

406. $12x^3 + \dfrac{2}{x^2}$

407. $\dfrac{x}{\sqrt{x^2+4}}$

408. $3x^2 - 12 = 0$, so $x = \pm 2$.

409. $x < -3$ or $x > 3$

410. $1 = 6 + c$, so $c = -5$.

411. $x^2 - x - 2 = 0$, so $x = 2$ or $x = -1$: the points $(2, 4)$ and $(-1, 1)$.

412. All real x except $x = 0$.

413. $\dfrac{3}{36} = \dfrac{1}{12}$

414. $27a^6b^3$

415. a) e^x b) $\dfrac{1}{x}$ c) $\cos x$ d) $-\operatorname{cosec}^2 x$ e) $(\ln 3)\,3^x$

416. a) Product b) Chain c) Quotient d) Term by term

417. $\dfrac{dy}{dx} = e^x \cos 2x - 2e^x \sin 2x = e^x(\cos 2x - 2\sin 2x)$

418. $\dfrac{dy}{dx} = 2xe^{3x} + 3x^2e^{3x} = xe^{3x}(2 + 3x)$

419. $\dfrac{dy}{dx} = 2xe^{-x} - x^2e^{-x} = xe^{-x}(2 - x)$

420. $\dfrac{dy}{dx} = e^{-3x} - 3xe^{-3x} = e^{-3x}(1 - 3x)$

421. $\dfrac{dy}{dx} = -e^{-x}\cos x - e^{-x}\sin x = -e^{-x}(\cos x + \sin x)$

422. $\dfrac{dy}{dx} = \dfrac{\frac{1}{x}e^x - e^x \ln x}{e^{2x}} = \dfrac{\frac{1}{x} - \ln x}{e^x} = \dfrac{1 - x\ln x}{xe^x}$

423. $\dfrac{dy}{dx} = \dfrac{e^{2x}\cos x - 2e^{2x}\sin x}{e^{4x}} = \dfrac{\cos x - 2\sin x}{e^{2x}}$

424. $\dfrac{dy}{dx} = \dfrac{2x\ln x - x^2 \cdot \frac{1}{x}}{(\ln x)^2} = \dfrac{x(2\ln x - 1)}{(\ln x)^2}$

425. $\dfrac{dy}{dx} = -e^x \sin(e^x)$

426. $\dfrac{dy}{dx} = \sec^2 x\, e^{\tan x}$

427. $\dfrac{dy}{dx} = \dfrac{e^x}{1 + e^x}$

428. $\dfrac{dy}{dx} = \dfrac{1 - \sin x}{x + \cos x}$

429. $\dfrac{dy}{dx} = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$

430. $\dfrac{dy}{dx} = \dfrac{2 + 2\cos 2x}{2x + \sin 2x}$

431. $\dfrac{dy}{dx} = 3\,(e^x + 1)^2 \times e^x = 3e^x\,(e^x + 1)^2$

432. $\dfrac{dy}{dx} = 2(\sin x + \cos x)(\cos x - \sin x)$

433. $y = (1+\ln x)^{\frac{1}{2}}$, so $\dfrac{dy}{dx} = \frac{1}{2}(1+\ln x)^{-\frac{1}{2}} \times \dfrac{1}{x} = \dfrac{1}{2x\sqrt{1+\ln x}}$.

434. $\dfrac{\cos x(1+\cos x)^2 + 2\sin^2 x(1+\cos x)}{(1+\cos x)^4} = \dfrac{\cos x + \cos^2 x + 2 - 2\cos^2 x}{(1+\cos x)^3} = \dfrac{(2-\cos x)(1+\cos x)}{(1+\cos x)^3} = \dfrac{2-\cos x}{(1+\cos x)^2}$

435. $\dfrac{dy}{dx} = \dfrac{\left(x^2+1\right)^2 - x\cdot 4x\left(x^2+1\right)}{\left(x^2+1\right)^4} = \dfrac{x^2+1-4x^2}{\left(x^2+1\right)^3} = \dfrac{1-3x^2}{\left(x^2+1\right)^3}$

436. $\dfrac{dy}{dx} = \dfrac{e^x\left(e^x+1\right)^3 - e^x\cdot 3e^x\left(e^x+1\right)^2}{\left(e^x+1\right)^6} = \dfrac{e^x\left(1-2e^x\right)}{\left(e^x+1\right)^4}$

437. $\dfrac{dy}{dx} = e^x(\sin x + \cos x) = 1$ at $x = 0$. The tangent is $y = x$.

438. Point $(1, \ln 2)$. $\dfrac{dy}{dx} = \dfrac{2x}{1+x^2} = 1$. $y = x - 1 + \ln 2$.

439. $\dfrac{dy}{dx} = e^{2x}(1+2x) = 1$ at $x = 0$, so the normal gradient is -1: $y = -x$.

440. In $e^x \sin x$ the last operation is multiplying two functions; in $e^{\sin x}$ one function is inside another. $\dfrac{d}{dx}e^x \sin x = e^x(\sin x + \cos x)$ and $\dfrac{d}{dx}e^{\sin x} = \cos x\, e^{\sin x}$.

441. $\dfrac{d}{dx}\ln(\sec x) = \dfrac{\tan x \sec x}{\sec x} = \tan x$

442. The student used the power rule on an exponential. For $e^{f(x)}$ the power stays the same and is multiplied by its derivative: $2xe^{x^2}$.

443. $\dfrac{dy}{dx} = \ln x + 1 - 1 = \ln x$, so the gradient at $x = e$ is 1.

444. a) $f'(x) = \dfrac{\frac{1}{x}\cdot x^2 - 2x\ln x}{x^4} = \dfrac{x(1-2\ln x)}{x^4} = \dfrac{1-2\ln x}{x^3}$ b) $\ln x = \frac{1}{2}$, so $x = \sqrt{e}$ and $f(\sqrt{e}) = \dfrac{\frac{1}{2}}{e} = \dfrac{1}{2e}$. The point is $\left(\sqrt{e}, \dfrac{1}{2e}\right)$.

445. $5e^{5x-1}$

446. $y = \ln 3 + 2\ln x$, so $\dfrac{dy}{dx} = \dfrac{2}{x}$.

447. $\dfrac{1}{(x+1)\ln 2}$

448. $(\ln 7)\, 7^x$

449. $2x\sec^2\left(x^2\right)$

450. $3\tan 3x \sec 3x$

451. $-\cot x \operatorname{cosec} x$

452. $-2\operatorname{cosec}^2 2x$

453. $2\sin 3x \times 3\cos 3x = 6\sin 3x \cos 3x$

454. $\cos x - x\sin x$

455. Point $(0,\ 1)$, $m = -1$: $y = -x + 1$.

456. Tangent gradient $\cos 0 = 1$, normal gradient -1: $y = -x$.

457. Quotient rule: $\dfrac{e^x(e^x+1) - (e^x-1)e^x}{(e^x+1)^2} = \dfrac{e^{2x} + e^x - e^{2x} + e^x}{(e^x+1)^2} = \dfrac{2e^x}{(e^x+1)^2}$

458. $-\dfrac{\sqrt{2}}{2}$

459. $\sqrt{2}$

460. $x = \dfrac{3\pi}{4}$ or $\dfrac{7\pi}{4}$

461. $2x = \dfrac{\pi}{2}$ or $\dfrac{3\pi}{2}$, so $x = \dfrac{\pi}{4}$ or $\dfrac{3\pi}{4}$.

462. $x = \ln 3$

463. $x = 25$

464. $\ln e^2 = 2$

465. $-2(3x+2)^{-3} \times 3 = -6(3x+2)^{-3}$

466. $\dfrac{2x(1-x) + x^2}{(1-x)^2} = \dfrac{2x - x^2}{(1-x)^2}$

467. Point $(-1,\ -2)$, $m = 2(-1) + 3 = 1$: $y = x - 1$.

468. $(2x-1)(x-3) = 0$, so $x = \frac{1}{2}$ or $x = 3$.

469. $x = 3$, $y = 9 - 18 + 5 = -4$: $(3,\ -4)$.

470. $3x - 1 = 5$ or $3x - 1 = -5$, so $x = 2$ or $x = -\frac{4}{3}$.

471. $x \geq 3$

472. $0.4 + 0.3 - 0.1 = 0.6$

473. $\dfrac{-4-4}{3+1} = -2$

474. $x^{-6} \times x^7 = x$